Physics

1. Ann is sitting on the edge of a carousel that has a radius of 6 m and is rotating steadily. Bob is standing still on the ground at a point that is 12 m from the centre of the carousel. At a particular instant, Bob observes Ann moving directly towards him with a speed of 1 ms^{-1}. With what speed does Ann observe Bob to be moving at that same moment?

2. A group of Alaskan gold prospectors reach a wide straight river that flows with uniform speed v. What immediately catches all their eyes is a huge gold nugget lying on the further bank, directly across the river. The laws governing prospecting in Alaska state that the first person to reach any particular place has the right to establish a mine there; speed is of the essence!

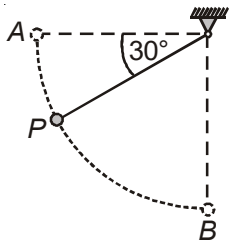
   Joe, one of the prospectors, has a canoe, which he can paddle in still water at the same speed u as he can hike along a river bank. What course of action should he take if u/v is (a) smaller than or (b) larger than a certain critical value? Assume that Joe first paddles across the river (in a straight line) and then, if necessary, hikes along the bank to reach the nugget.

3. A boy is running north, with a speed of $v = 5\ ms^{-1}$, on the smooth ice cover of a large frozen lake. The coefficient of friction (both kinetic and static) between the tread of his trainers and the ice is $\mu = 0.1$. For the sake of simplicity, assume that the normal force he exerts on the ice, which in reality changes with time, can be substituted by its average value.

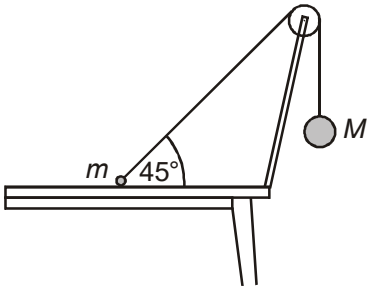
   (a) What is the minimal time that he needs to change direction, so that he is running east with the same speed v?

   (b) Find the boy's trajectory during the turn in this optimal case.

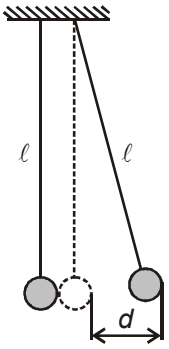
4. A simple pendulum is released from rest with its string horizontal. Which of the two arcs, AP and PB as defined in the figure, will its bob cover in a shorter time?



5. Two spherical bodies, with masses m and M, are joined together by a light thread that passes over a table-mounted pulley of negligible mass. Initially they are held in the positions shown in the figure; then, at a given moment, both of them are released. Mass M is many times – say, one thousand times – larger than mass m. The friction between the smaller ball and the surface of the table is negligible. Will the lighter ball be lifted from the table-top immediately after the release?

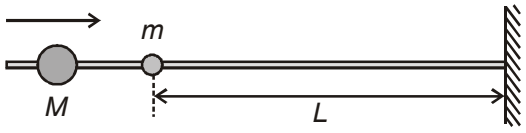


6. Two identical balls are suspended on (vertical) threads of length ℓ so as to just touch each other and, as shown in the figure, can swing in the plane defined by their suspension points and their centres. One of the balls is drawn aside in this plane through a distance d ($<< \ell$) and then released. Each time the balls collide, they do so inelastically, and as measured in their centre-of-mass frame, their velocities decrease by a factor of k, the coefficient of restitution, where $0 < k < 1$.



How will the balls be moving at a much later time? What will be the amplitude of the swings after a large number of collisions have taken place? Assume that the damping due to air resistance is very small.

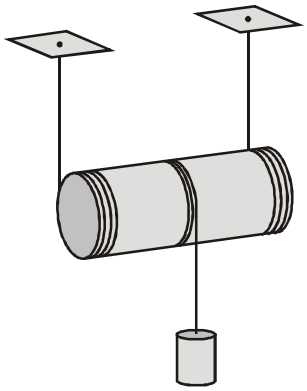
7. Two small balls are threaded onto a frictionless horizontal rod that protrudes from a vertical wall. The lighter ball with mass m is initially at rest, a distance L from the wall, while the much heavier second ball, of mass M, approaches the wall from a distance greater than L (see figure). After their elastic collision, the ball of mass m slides towards the wall, bounces back elastically, and again collides with the heavier ball. The process then repeats itself, over and over.



How close does the heavier ball get to the wall?

Consider the balls as point-like particles, and where appropriate, assume that $m << M$.

8. A cylindrically symmetric (but not necessarily homogeneous) body is attached to two identical cords at points near its ends. The cords are partially wound in the same sense around the cylinder, and their free ends are fastened to points on a ceiling; initially, the cords are vertical and the cylinder is horizontal. A third cord is attached to and wound (in the same sense as the other two cords) around the middle of the cylinder; a heavy weight is tied to the free end of this cord (see figure).

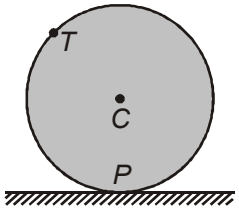


When the system is released from rest, what is the acceleration of the heavy weight?

9. A billiard ball, initially at rest on a billiard table, is struck by a cue tip at the point *T* shown in the figure. The cue lies in the vertical plane containing *T*, the centre *C* of the ball and the ball's point of contact *P* with the table; consequently, so does the line of action of the resulting impulse. Find the direction in which the cue should be aligned in order that, after the shot,

   (a) the ball's subsequent rotational and slipping motions terminate at the same instant, and the ball comes to a halt,

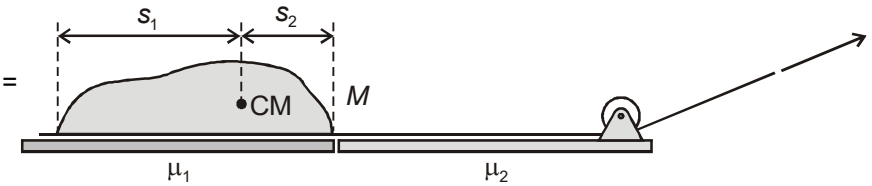
   (b) the ball rolls without slipping, whatever the value of the coefficient of static friction between it and the table.



Assume that, as a result of chalking the cue tip, the coefficient of friction between it and the ball is sufficiently large that there is no slippage between them during the cue stroke.

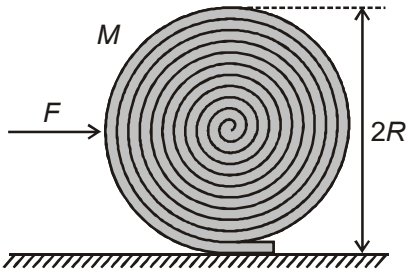
10. A sack, full of sand and of mass *M*, lies on a carpet, which, in turn, lies on a relatively smooth horizontal surface. The distribution of the sack's weight is not uniform, but it is known that its centre of mass (CM) lies at distances s_1 and s_2 from its extremities.

    By pulling horizontally on the carpet, both it and the sack are to be moved onto an immediately adjacent rougher surface at the same height (see figure). How much work is required to do this if the coefficients of friction between the carpet and the original and final surfaces are μ_1 and μ_2, respectively?



11. The cable strung between two neighbouring electricity posts sags a little. The mass per unit length of the cable is λ, the distance between the posts is *L*, and the 'maximum sag' of the cable is *d* (*d* << *L*). What is the (approximate) tension in it?

12. A flexible scatter rug (also known as a runner), of mass M and length L, is tightly coiled into a cylinder of radius R (<< L), as in the figure. If the coiled rug is released, then, in the absence of rolling friction, it spontaneously unrolls to its full length.



(a) Explain in terms of the forces involved why this happens.

(b) How large a horizontal force F, applied as in the figure, is required to prevent the unrolling of the rug?

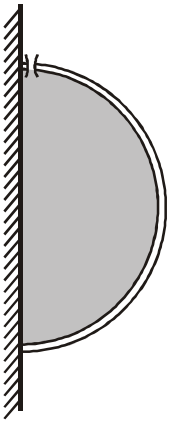
13. The two ends of a 40 cm long chain are fixed at the same height, as shown in the figure. Find the radius of curvature of the chain

(a) at its lowest point and

(b) at the suspension points.



14. A thin-walled hemispherical shell of mass m and radius R is pressed against a smooth vertical wall, and through a small aperture at its top, is filled with a liquid (say, water) of density ρ (see figure).What are the minimal magnitude and the direction of the force that has to be applied to the shell if liquid is not to escape from it? To which point on the shell should this force be applied?



15. A razor-blade floats on the surface of water contained in a glass. When the glass is gently shaken, the razor-blade sinks. How, if at all, does the water level change as a result?

16. Two spherical soap bubbles with different initial radii coagulate, and the radii of their free surfaces after the coagulation are R and r. What is the radius of curvature of the soap film that separates the two bubbles? What is the radius of this film's circular perimeter?

17. You are given 1 kg of distilled water at 0°C, an equal mass of boiled tap water at 100°C, and the task of warming the distilled water to 60°C. How would you set about it?

No additional water, hot or cold, is available, but you are provided with both insulating and conducting materials (in sheet form) and a suitable selection of tools.

What is the maximum possible temperature, in principle, to which your method could warm the distilled water?

18. A quantity, n mol say, of helium gas undergoes a thermodynamic process (neither isothermal nor isobaric) in which the molar heat capacity C of the gas can be described as a function of the absolute temperature by $C = \frac{3RT}{(4T_0)}$, where R is the gas constant and T_0 is the initial temperature of the helium gas. Find the work done on the system up until the point at which the helium gas reaches its minimal volume.

19. The three vertices of a metal plate in the form of an equilateral triangle are held at constant temperatures T_1, T_2 and T_3 (by cooling or heating, as necessary). What temperature do you expect at the centre of the triangle? Prove that your answer is correct.

20. If 1 kg of water freezes, then 334 kJ of energy are released. How much energy is released if 1 kg of supercooled water at –10°C freezes, and during the freezing process the temperature remains constant?

21. Water is boiling in a narrow test tube that is open at the top. Just before the last few drops are vaporised, the tube is rapidly and hermetically sealed. The temperature at the top of the tube is then slowly increased to 200°C, while the very bottom of it is maintained at 100°C.

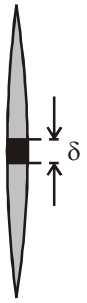
    What is the pressure of the steam in the test tube?

22. On a spherical planet, the refractive index of the atmosphere, as a function of altitude h above the surface, varies according to the formula

$$n(h) = \frac{n_0}{1 + \varepsilon h},$$

    where n_0 and ε are constants. Curiously, any laser beam, directed horizontally, but at an arbitrary altitude, follows a trajectory that circles the planet. What is the radius of the planet?

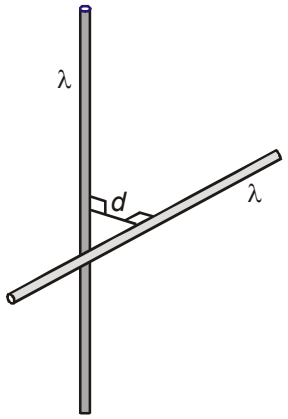
23. A converging lens of focal length f is cut along a plane that contains the optical axis of the lens, and a small, black plate of thickness δ is placed between the two half-lenses. A point-like source emitting monochromatic light of wavelength λ is located on the 'optical axis', a distance p from the lens ($p > f$).



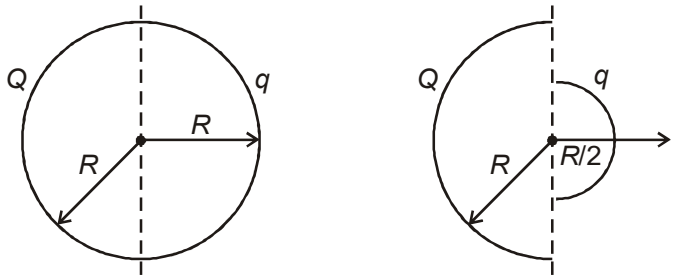
    How many interference fringes can be seen on a screen placed a distance H behind the lens, with its plane perpendicular to the optical axis?

    Data: f = 10 cm, p = 20 cm, δ = 1 mm, λ = 0.5 μm, H = 50 cm.

24. Two very long, thin, insulating rods, each carrying uniform linear charge density λ, lie in perpendicular directions (in three-dimensional space), a distance d from each other (see figure). What is the magnitude of the force of repulsion between them?



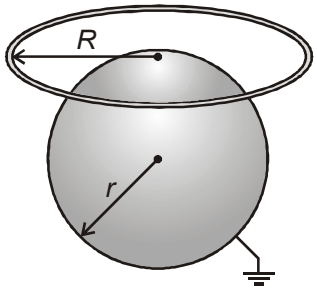
25. Two insulating hemispherical shells (e.g. the two halves of a ping-pong ball) are placed very close to each other with their centres almost coincident, as shown in figure (a). Both of them are uniformly charged, one with total electric charge Q, and the other with total charge q.



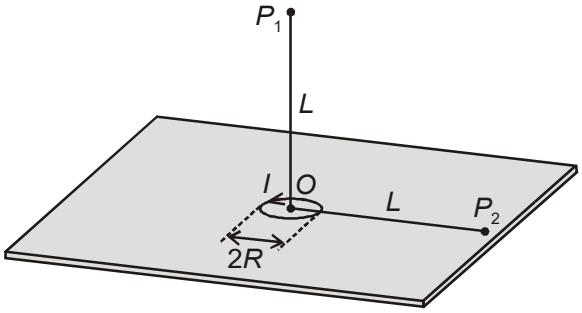
(a) What is the magnitude of the net electric force exerted by the two bodies on each other?

(b) Does the result change if the radius of one of the hemispherical shells is only one-half that of the other, as shown in figure (b)?

26. A thin metal ring of radius R is charged in such a way that the electric potential at its centre is V_0. The ring is now placed horizontally above a grounded solid metal sphere of radius r, so that the centre of the ring and the top of the sphere coincide. Find the total charge induced on the sphere.



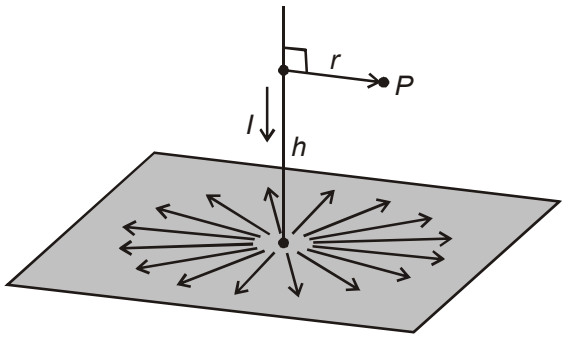
27. A circular wire loop of radius R, and carrying an electric current I, is at rest in a horizontal plane (see figure). Find the magnetic field strength at a distance L (>> R) from the centre O of the circular loop:



(a) at a point, P_1, situated on the vertical axis passing through O,

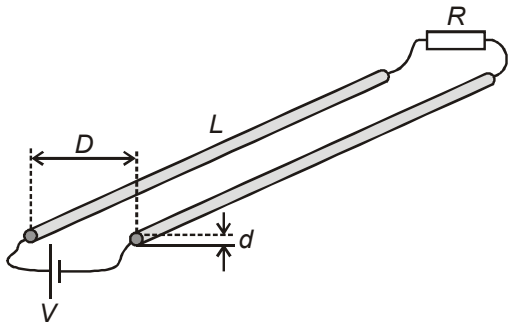
(b) at a point, P_2 in the plane of the current loop.

28. One end of a long, straight, current-carrying metal wire is electrically connected to a very large, thin, homogeneous metal plate, whose plane is perpendicular to the wire, and whose distant perimeter is earthed. One of the terminals of the battery supplying the current is also earthed.



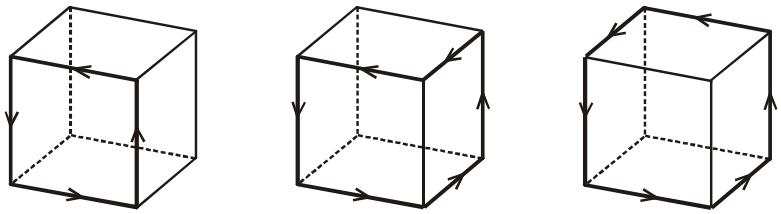
Ignoring the Earth's field, find the magnitude and direction of the magnetic field at a point P that is a distance r from the wire and at a height h above the metal plate, when a steady current I flows in the wire (see figure). Describe the magnetic field above and below the plate.

29. Two identical, very long, cylindrical conductors, of diameter d and negligible resistance, are placed parallel to each other with their axes separated by $D = 50d$. A battery of electromotive force (voltage) V is connected between the left-hand ends of the wires, while a resistor with resistance R is connected across their other ends (see figure). Find the resistance R that makes the electrical and magnetic forces between the conductors equal.



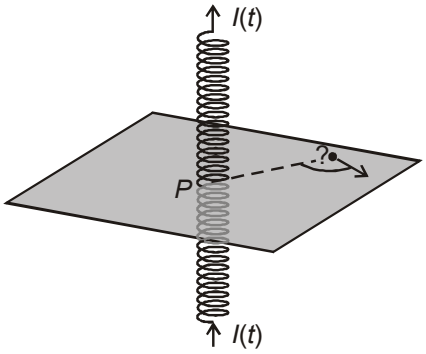
30. Two identical superconducting rings are very far from each other. The current in one of the rings (say, A) is I_0, but there is no current in the other (B). The two rings are now slowly brought closer together. Find the current that flows in A when that in B has increased to I_1.

31. Three metal wires of lengths $4a$, $6a$ and $6a$ are arranged along the edges of a cube of side a in three different ways, as shown in the figure. The coefficient of self-inductance of the square shown in figure (a) is measured as L_1, and for the arrangement shown in figure (b) it is L_2. How large, expressed in terms of L_1 and L_2, would it be for the arrangement shown in figure (c)?



(a) (b) (c)

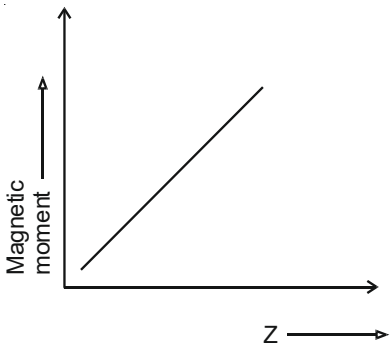
32. A small, electrically charged pearl, initially at rest, can move on a horizontal, frictionless plane. Not far from the pearl, there is a long, vertical solenoid, in which the electric current is first increased uniformly from zero to a given value, and then uniformly decreased back to zero. In which direction, relative to point *P* in the figure, will the pearl have moved by the end of the process? Or will it (still) be where it started?

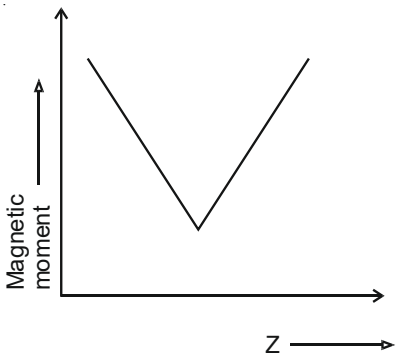


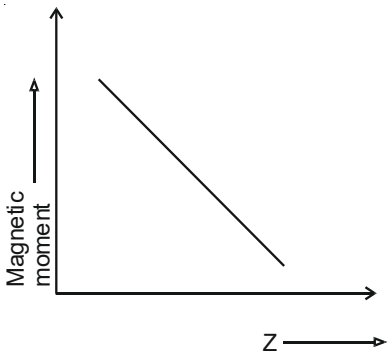
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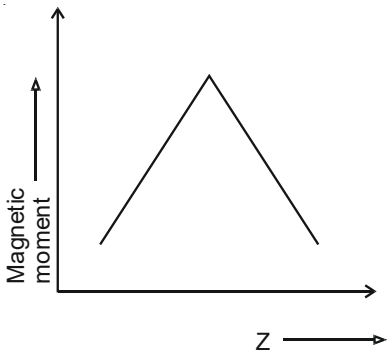
Chemistry

1. Which of the following graph is correct representation between atomic number (Z) and magnetic moment of *d*-block elements? [Outer electronic configuration : $(n-1)d^x ns^{1 \text{ or } 2}$]

(A) 

(B) 

(C) 

(D) 

2. Which of the following transformation least energy is required?

(A) $F^-_{(g)} \rightarrow F_{(g)} + e^-$

(B) $P^-_{(g)} \rightarrow P_{(g)} + e^-$

(C) $S^-_{(g)} \rightarrow S_{(g)} + e^-$

(D) $Cl^-_{(g)} \rightarrow Cl_{(g)} + e^-$

3. Which of the following order is incorrect?

(A) Electronegativity of central atom : $CF_4 > CH_4 > SiH_4$

(B) Hydration energy : $Al^{3+} > Be^{2+} > Mg^{2+} > Na^+$

(C) Electrical conductance : $F^-_{(aq.)} > Cl^-_{(aq.)} > S^{2-}_{(aq.)}$

(D) Magnetic moment : $Ni^{4+} > V^{3+} > Sr^{2+}$

4. The ground state electronic configurations of the elements, U, V, W, X and Y (these symbols do not have any chemical significance) are as follows:

U $1s^2 2s^2 2p^3$

V $1s^2 2s^2 2p^6 3s^1$

W $1s^2 2s^2 2p^6 3s^2 3p^2$

X $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$

Y $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$

Determine which sequence of elements satisfy the following statement :

(i) Element forms a carbonate which is not decomposed by heating

(ii) Element is most likely to form coloured ionic compounds

(iii) Element has larger atomic radius

(iv) Element forms only acidic oxide

(A) V W Y U (B) V X Y W

(C) V W Y X (D) V X W U

5. A, B and C are hydroxy-compounds of the elements X, Y and Z respectively. X, Y and Z are in the same period of the periodic table. A gives an aqueous solution of pH less than seven. B reacts with both strong acids and strong alkalis. C gives an aqueous solution which is strongly alkaline.

Which of the following statements is/are true?

(i) The three elements are metals.

(ii) The electronegativities decrease from X to Y to Z.

(iii) The atomic radius decreases in the order X, Y and Z.

(iv) X, Y and Z could be phosphorus, aluminium and sodium respectively.

(A) Only i, ii, iii correct (B) Only i, iii correct

(C) Only ii, iv correct (D) Only ii, iii, iv correct

6. Consider the following ionisation reactions:

I. E. (kJ mol^{-1}) I.E. (kJ mol^{-1})

$A_{(g)} \longrightarrow A^+_{(g)} + e^-$, A_1 $B_{(g)} \longrightarrow B^+_{(g)} + e^-$, B_1

$B^+_{(g)} \longrightarrow B^{2+}_{(g)} + e^-$, B_2 $C_{(g)} \longrightarrow C^+_{(g)} + e^-$, C_1

$C^+_{(g)} \longrightarrow C^{2+}_{(g)} + e^-$, C_2 $C^{2+}_{(g)} \longrightarrow C^{3+}_{(g)} + e^-, C_3$

If monovalent positive ion of A, divalent positive ion of B and trivalent positive ion of C have zero electron. Then incorrect order of corresponding I.E. is:

(A) $C_3 > B_2 > A_1$

(B) $B_1 > A_1 > C_1$

(C) $C_3 > C_2 > B_2$

(D) $B_2 > C_3 > A_1$

7. First three ionisation energies (in kJ/mol) of three representative elements are given below:

Element	IE_1	IE_2	IE_3
P	495.8	4562	6910
Q	737.7	1451	7733
R	577.5	1817	2745

Then incorrect option is :

(A) Q : Alkaline earth metal

(B) P : Alkali metals

(C) R : s-block element

(D) They belong to same period

8. Aqueous solutions of two compounds $M_1 - O - H$ and $M_2 - O - H$ are prepared in two different beakers. If, the electronegativity of M_1 = 3.4, M_2 = 1.2, O = 3.5 and H = 2.1, then the nature of two solutions will be respectively:

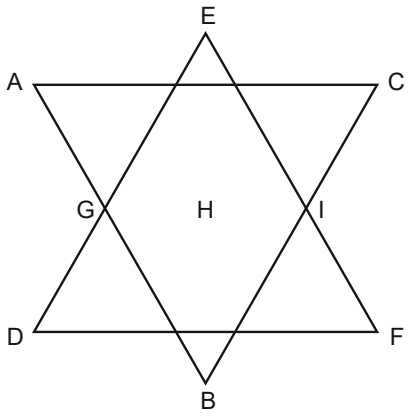
(A) Acidic, basic

(B) Acidic, acidic

(C) Basic, acidic

(D) Basic, basic

Paragraph for Q. Nos. 9 to 11



There are nine elements A to I. These are belongs to p-block element other than halogen. If atomic number of B is average of atomic number of A and C and atomic number of E is average of atomic number of D and F and atomic number of H is average of atomic number of G and I. Atomic numbers of B, E and H are 7, 15 and 83 respectively and atomic numbers of C, I and F are greater than A, G and D respectively.

9. The incorrect order is :

(A) F > E : Second ionisation energy

(B) B > C : Z_{eff} on valence shell

(C) I > H : First ionisation energy

(D) C > F > E : Electronegativity

10. The correct statement is :

(A) +5 oxidation state of H is more stable than its +3 oxidation state.

(B) G^{2+} is better oxidising agent than G^{4+}

(C) +3 oxidation state of E is more stable than its +5 oxidation state

(D) Ionisation energy of G is greater than that of 'Tin',

11. Which of the following statement is incorrect?

(A) B_2C_5 is acidic in nature

(B) AC_2 is acidic in nature

(C) FC_3 is basic in nature

(D) GC_2 is amphoteric in nature

12. S, T and U are the aqueous chlorides of the elements X, Y and Z respectively. X, Y and Z are in the same period of the periodic table. U gives a white precipitate with NaOH but this white precipitate dissolves as more NaOH is added. When NaOH is added to T, a white precipitate forms which does not dissolve when more base is added. S does not give precipitate with NaOH.

Which of the following statements are correct?

(A) The three elements are metals

(B) The electronegativity decreases from X to Y to Z

(C) X, Y and Z could be sodium, magnesium and aluminium respectively

(D) The first ionization increases from X to Y to Z

13. There are three elements A, B, and C. Their atomic number are Z_1, Z_2 and Z_3 respectively. If $Z_1 - Z_2 = 2$ and $\frac{Z_1 + Z_2}{2} = Z_3 - 2$ and the electronic configuration of element A is $[Ar]3d^6 4s^2$, then correct order of magnetic momentum is/are:

(A) $B^+ > A^{2+} > C^{2+}$ (B) $A^{3+} > B^{2+} > C$

(C) $B > A > C^{2+}$ (D) $B = A^{3+} > C^{3+}$

14. Consider the value of all four quantum number for last electron and spin multiplicity (2s + 1) for given two element X and Y in their ground state:

	n	l	m	s	|2s + 1|
X :	2	0	0	+1/2	1
Y :	2	1	–1	–1/2	4

The according to given information the correct statement is :

(A) The bond angle (H — Y — H) of possible hydride of element Y is less than 109°28′

(B) The possible halide of X has two vacant p-orbitals on its central atom

(C) Magnetic moment of Y is greater than X

(D) X and Y element exhibits only single oxidation state

15. Consider the following values of I.E. (eV) for elements W and X:

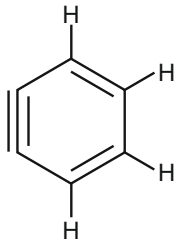
Element	$I.E._1$	$I.E._2$	$I.E._3$	$I.E._4$
W	10.5	15.5	24.9	79.8
X	8	14.8	78.9	105.8

Other two element Y and Z have outer electronic configuration ns^2np^4 and ns^2np^5 respectively. Then according to given information which of the following compound(s) is/are not possible?

(A) W_2Y_3 (B) X_2Y_3

(C) WZ_2 (D) XZ_2

16. How many sp^2 and *sp*-hybridised carbon atoms are present respectively in the following compound?



(A) 4, 2 (B) 6, 0

(C) 3, 3 (D) 5, 1

17. Give the correct order of initials **T** or **F** for following statements. Use **T** if statement is true and **F** if it is false:

(i) The order of repulsion between different pair of electrons is $l_p - l_p > l_p - b_p > b_p - b_p$

(ii) In general, as the number of lone pair of electrons on central atom increases, value of bond angle from normal bond angle also increases

(iii) The number of lone pair on O in H_2O is 2 while on N in NH_3 is 1

(iv) The structures of xenon fluorides and xenon oxyfluorides could not be explained on the basis of VSEPR theory

(A) T T T F (B) T F T F

(C) T F T T (D) T F F F

18. Which of the following overlaps is incorrect (assuming Z-axis is internuclear axis)?

(i) $2p_y + 2p_y \rightarrow \pi$-Bond formation

(ii) $2p_x + 2p_x \rightarrow \sigma$-Bond formation

(iii) $3d_{xy} + 3d_{xy} \rightarrow \pi$-Bond formation

(iv) $2s + 2p_y \rightarrow \pi$-Bond formation

(v) $3d_{xy} + 3d_{xy} \rightarrow \delta$-Bond formation

(vi) $2p_z + 2p_z \rightarrow \sigma$-Bond formation

(A) i, ii, iii (B) iii, vi

(C) ii, v (D) ii, iii, iv

19. Choose the correct code of characteristics for the given order of hybrid orbitals of same atom,

$sp < sp^2 < sp^3$

(i) Electronegativity

(ii) Bond angle between same hybrid orbitals

(iii) Size

(iv) Energy level

(A) (ii), (iii) and (iv)

(B) (iii), (iv)

(C) (ii) and (iv)

(D) (i), (ii), (iii) and (iv)

20. Which of the following is incorrectly match?

	Hybridisation	**Geometry**	**Orbitals use**
(A)	sp^3d	Trigonal bipyramidal	$s + p_x + p_y + p_z + d_{z^2}$
(B)	sp^3d^3	Pentagonal bipyramidal	$s + p_x + p_y + p_z + d_{x^2-y^2} + d_{z^2} + d_{xy}$
(C)	sp^3d^2	Capped octahedral	$s + p_x + p_y + p_z + d_{x^2-y^2} + d_{z^2}$
(D)	sp^3	Tetrahedral	$s + p_x + p_y + p_z$

Paragraph for Q. No. 21

According to VBT any covalent bond will be formed by overlapping of atomic orbitals of bonded atoms provided atomic orbitals must be half-filled and electrons be in opposite spin. According to type of overlapping covalent bonds can be classified as (a) σ-bond (b) π-bond (C) δ-bond.

21. If $F_2C_1 = C_2$ part of $F_2C_1 = C_2 = C_3 = C_4F_2$ lies in yz-plane, then incorrect statement is

(A) Nodal plane of π-bond between C_1 and C_2 lies in yz-plane, formed by sideways overlapping of p_x-orbitals

(B) Nodal plane of π-bond between C_2 and C_3 lies in xz-plane, formed by sideways overlapping of p_y-orbitals

(C) Nodal plane of π-bond between C_3 and C_4 lies in yz-plane, formed by sideways overlapping of p_y-orbitals

(D) Nodal plane of π-bond between C_2 and C_3 lies in xy-plane, formed by sideways overlapping of p_z-orbitals

22. Consider the following molecule :

$$\underset{(1)}{H_2C} = \underset{(2)}{C} = \underset{(3)}{C} = \underset{(4)}{C} = \underset{(5)}{CF_2}$$

If hybridization of $C_{(1)}$ carbon atom is $sp^2(s + p_y + p_z)$ and hybridization of $C_{(4)}$ carbon atom is $sp(s + p_z)$. Then according to given information the **correct** statement(s) is/are:

(A) Nodal plane of π-bond between $C_{(2)}$ and $C_{(3)}$ lies in xz-plane, formed by sideways overlapping of p_y-orbitals

(B) Nodal plane of π-bond between $C_{(3)}$ and $C_{(4)}$ lies in yz-plane, formed by side ways overlapping of p_x-orbitals

(C) The orbitals involve in hybridization of $C_{(5)}$ carbon atom are $s + p_x + p_z$

(D) Nodal plane of π-bond between $C_{(1)}$ and $C_{(2)}$ lies in yz-plane, formed by side ways overlapping of p_y-orbitals

23. Consider the following two molecules and according to the given information select correct statement(s) about AX_2 and AY_2 :

Where A : 16th group of 3rd period element

X : More electronegative than (A) and same group number of (A)

Y : Less atomic size than (A) and same period number of (A)

(A) The hybridization of central atoms are different in both compounds

(B) The shape of both molecules are same

(C) Both compounds are planar

(D) The X — A — X bond angle is less than Y — A — Y bond angle

24. Which of the following combination of bond pair (b.p.) and lone pair (l.p.) give same shape?

(i) 3 b.p. + 1 l.p.

(ii) 2 b.p. + 2 l.p.

(iii) 2 b.p. + 1 l.p.

(iv) 2 b.p. + 0 l.p.

(v) 3 b.p. + 2 l.p.

(vi) 2 b.p. + 3 l.p.

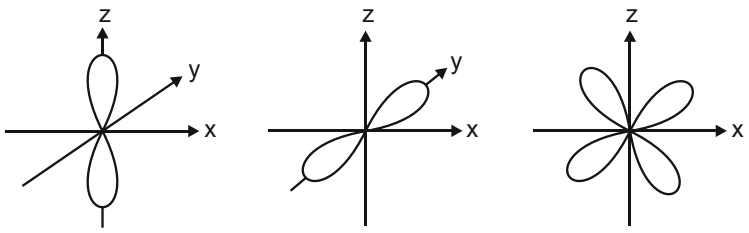
(A) (ii) and (iii)

(B) (iv) and (v)

(C) (iv) and (vi)

(D) (iii) and (vi)

25. Consider the following three orbitals :



(i) (ii) (iii)

Correct statement(s) regarding given information is/are:

(A) Orbitals (i) and (ii) can never form any type of covalent bond

(B) If internuclear axis is x, then combination of (ii) and (iii) orbitals can form π-bond

(C) Orbital (iii) can form δ-bond with other orbital having same orientation of lobes

(D) If internuclear axis is x, then combination of (i) and (iii) orbitals can form π-bond

26.

Column-I (Type of bond formed)	**Column-II** [Combining orbitals, (internuclear axis)]
(A) π-bond	(P) $d_{yz} + p_y$, (z)
(B) σ-bond	(Q) $s + p_x$, (y)
(C) δ-bond	(R) $d_{yz} + d_{yz}$, (x)
(D) Non-bonding	(S) s + s, (z)
	(T) $s + d_{xy}$, (y)

27. Consider the following five groups (According to modern periodic table) of elements with their increasing order of atomic numbers :

Group 1 $\rightarrow$ A, B, C, D, E

Group 2 $\rightarrow$ F, G, H, I, J

Group 13 $\rightarrow$ K, L, M, N, O

Group 15 $\rightarrow$ P, Q, R, S, T

Group 17 $\rightarrow$ U, V, W, X, Y

If first and last element of each group belongs to 2nd and 6th period respectively and Z represents to carbonate ion $\left(CO_3^{2-}\right)$ then consider the following orders.

(i) $O^+ > H^{2+}$; Polarising power

(ii) $T^{3+} > S^{3+} > R^{3+}$; Stability of cation

(iii) $U^-(aq) > V^-(aq) > W^-(aq) > X^-(aq)$; size

(iv) $JV_2 < IV_2 < GV_2 < LV_3$; Covalent character

(v) GZ > IZ > JZ; Thermal stability

(vi) AV > BV > CV > DV > EV; Thermal stability

(vii) $C_3P > B_3P > A_3P$; Lattice energy

(viii) $KU_3 < KV_3 < KW_3 < KX_3$; Melting point

Then calculate value of $|p - q|^2$, here p and q are correct and incorrect orders in the given eight orders respectively.

28. Consider the following orbitals:

(i) $3p_x$

(ii) $4d_{z^2}$

(iii) $3d_{x^2-y^2}$

(iv) $3d_{yz}$

Then calculate value of x + y – z here x is total number of gerade orbital and y is total number of ungerade orbitals and z is total number of axial orbital in given above orbitals.

29. O_2F_2 is an unstable yellow orange solid and H_2O_2 is a colourless liquid both have O — O bond and O — O bond length in H_2O_2 and O_2F_2 respectively is :

(A) 1.22 Å, 1.48 Å

(B) 1.48 Å, 1.22, Å

(C) 1.22 Å, 1.22 Å

(D) 1.48 Å, 1.48 Å

30. Solubility of alkali metal fluorides increases down the group. Select correct explanation for given statement :

(A) Hydration energy increases and lattice energy decreases down the group

(B) Both energy decrease down the group but decrease in hydration energy is rapid

(C) Both energy decreases down the group but decrease in lattice energy is rapid

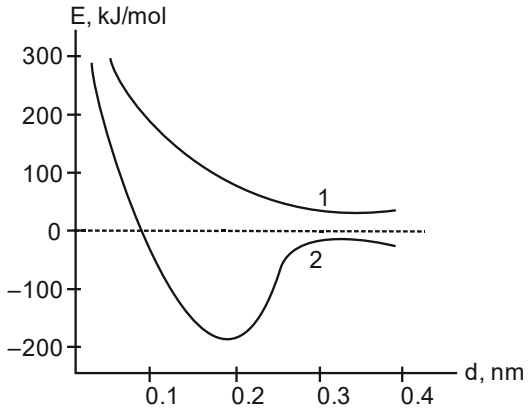
(D) Both energy increase down the group but increase in hydration energy is rapid

31. A molecule may be represented by three structures having energies E_1, E_2 and E_3, respectively. The energies of these structures follow the order $E_3 < E_2 < E_1$, respectively. If the experimental bond energy of the molecule is E_0, the resonance energy is :

(A) $(E_1 + E_2 + E_3) - E_0$ (B) $E_0 - E_3$

(C) $E_0 - E_1$ (D) $E_0 - E_2$

32. Consider the given figure showing the formation of H_2^+ ion depending on internuclear distance *versus* potential energy of the system.



Which is correct statement ?

(A) Curve-1 represents the most stable state of the system for H_2^+ ion

(B) Curve-2 represents the most stable state of the system for H_2^+ ion

(C) Curve-1 indicates that the molecular hydrogen ion is formed

(D) Curve-2 represents the energy level of the antibonding region

33. The incorrect statement(s) regarding $\dot{C}X_3$ species is :

(A) If electronegative of surrounding element X is less than 2.5, then central carbon atom used almost 33% s-character in their hybrid bonding orbitals

(B) If electronegativity of surrounding element X is greater than 2.5, then central carbon atom used almost 25% s-character in their hybrid bonding orbitals

(C) If X is F, then species should be polar and pyramidal

(D) If X is H, then species should be polar and planar

34. The incorrect statement about carbene (CH_2) is

(A) In singlet carbene, carbon is sp^2-hybridized

(B) Triplet carbene is less stable than singlet carbene

(C) Stability order of singlet halocarbenes is CHF > CHCl > CHBr

(D) Triplet Carbene is more stable than singlet carbene

35. Consider the following reaction :

$MX_4 + X'_2 \longrightarrow MX_4X'_2$

If atomic number of M is 52 and X and X′ are halogens and X′ is more electronegative than X. Then choose correct statement regarding given information:

(A) Both X′ atoms occupy axial positions which are formed by overlapping of p and d-orbitals only

(B) All M — X bond lengths are identical in both MX_4 and $MX_4\ X'_2$ compounds

(C) Central atom M does not use anyone valence non-axial set of d-orbital in hybridization of final product

(D) Hybridization of central atom M remains same in both reactant and final product

36. Consider the following information (X = F or Cl)

Molecule	P — X (axial) bond length	P — X (Equatorial) bond length
PF_5	a	b
PF_4CH_3	c	d
$PF_3(CH_3)_2$	e	f
PCl_5	g	h

According to given information choose the incorrect order of bond length:

(A) g > a > d > b (B) g > e > f > b

(C) f > d > a > b (D) c > f > d > b

37. Select the incorrect statement about N_2F_4 and N_2H_4 :

(i) In N_2F_4, d-orbitals are contracted by electronegative fluorine atoms, but d-orbital contraction is not possible by H-atom in N_2H_4

(ii) The N — N bond energy in N_2F_4 is more than N — N bond energy in N_2H_4

(iii) The N — N bond length in N_2F_4 is more than that of in N_2H_4

(iv) The N — N bond length in N_2F_4 is less than that of in N_2H_4

Choose the correct code :

(A) i, ii and iii (B) i and iii

(C) ii and iv (D) ii and iii

38. Cis-butene dioic acid $\xrightleftharpoons{Ka_1(-H^+)} X_1^- \xrightleftharpoons{Ka_2(-H^+)} X_2^{2-}$

Trans-butene dioic acid $\xrightleftharpoons{Ka'_1(-H^+)} Y_1^- \xrightleftharpoons{Ka'_2(-H^+)} Y_2^{2-}$

The incorrect statement regarding above information is :

(A) X_2^{2-} species is more basic than Y_2^{2-} species

(B) X_1^- species is more basic than Y_1^- species

(C) K_{a_1} is greater than K'_{a_1}

(D) K'_{a_2} is greater than K_{a_2}

39. The incorrect statement regarding $O(SiH_3)_2$ and OCl_2 molecule is/are :

(A) The strength of back bonding is more in $O(SiH_3)_2$ molecule than OCl_2 molecule

(B) Si — O — Si bond angle in $O(SiH_3)_2$ is greater than Cl — O — Cl bond angle in OCl_2

(C) The nature of back bond in both molecules is $2p_\pi$-$3d_\pi$

(D) Hybridisation of central O-atom in both molecules is same

40. Hybridisation of central atom does not always change due to back bonding. This statement is valid for which of the following compounds?

(i) CCl_3^-

(ii) BF_3

(iii) $(SiH_3)_2O$

(iv) $N(SiH_3)_3$

(A) (i), (ii)

(B) (i), (iii)

(C) (ii), (iii)

(D) All

41. The incorrect statement regarding molecular orbital(s) is

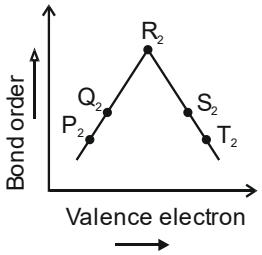
(A) If there is a nodal plane perpendicular to the internuclear axis and lying between the nuclei of bonded atoms then corresponding orbitals is antibonding M.O.

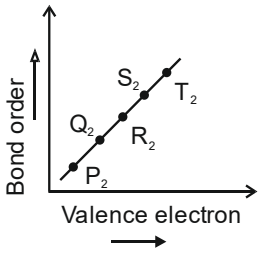
(B) If a nodal plane lies in the inter-nuclear axis, then corresponding orbitals is pi(π) bonding M.O

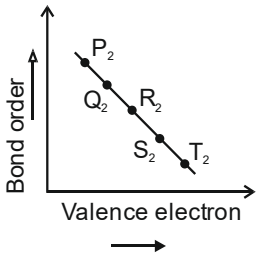
(C) The σ-bonding molecular orbital does not contain nodal planes containing the internuclear axis

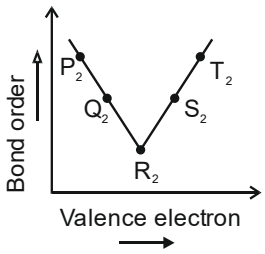
(D) The δ-bonding molecular orbital possesses three nodal planes containing the internuclear axis

42. If P to T are second period p-block elements then which of the following graph show correct relation between valence electrons in P_2 to T_2 (corresponding molecules) and their bond order?

(A)


(B)


(C)


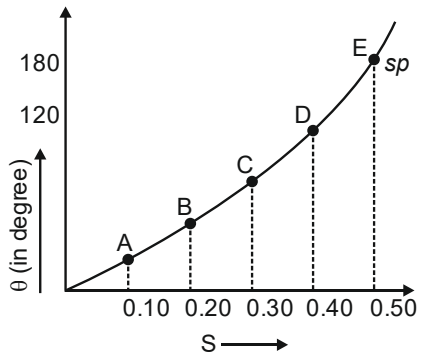
(D)


Paragraph for Q. Nos. 43 to 45

Compound, MX_n type (n = 2 or 3 or 4)	Value of cosθ (θ = bond angle between equivalent hybrid orbitals)
P	–0.241
Q	–0.292
R	–0.5
S	–0.325
T	–0.469

In all expected compounds each central atom only uses its s and p-orbitals in hybridization. The relationship between bond angle θ and decimal fraction of s and p-character present in the equivalent hybrid orbitals is given by :

$\cos\theta = \frac{S}{S-1} = \frac{P-1}{P}, S =$ decimal fraction of s-character in the equivalent hybrid orbital P = decimal fraction of p-character in the equivalent hybrid orbital.



43. The correct order of % p-character in bond pairs of central atoms in the following compounds is

(A) P > T > S > Q > R

(B) S > R > T > P > Q

(C) P > Q > S > R > T

(D) P > Q > S > T > R

44. If the value of n is 2 for compound T, then number of lone pair present at central atom of compound T will be

(A) 0 (B) 1

(C) 2 (D) 3

45. The incorrect statement is

(A) The ratio of % p-character to % s-character is less than four, for the bond pair of central atom of compound S

(B) Central atom uses three hybrid orbitals to form compound R

(C) Central atom uses four hybrid orbitals to form compound S

(D) There are three compounds present between point C to E, according to % s-character in bond pair of central atom

46. Two compounds PX_2Y_3 and PX_3Y_2. (Where P = Phosphorous atom and X, Y = monovalent atoms). If all X atoms are replaced by Z atoms and electronegativity order is X > Y > Z.

Then incorrect statement(s) is/are

(A) The dipole moment of product obtained from PX_2Y_3 is non-zero

(B) The dipole moment of product obtained from PX_2Y_3 is zero

(C) The dipole moment of product obtained from PX_3Y_2 is zero

(D) The dipole moment of product obtained from PX_3Y_2 is non-zero

47. Consider the following three compounds (i) AX_{2n}^{n-}, (ii) AX_{3n} and (iii) AX_{4n}^{n+}, where central atom A is 15th group element and their maximum covalency is 3n. If total number of proton in surrounding atom X is n and value of n is one, then calculate value of "$a^3 + b^2 + c$". (Where a, b and c are total number of lone pair at central atom in compound (i), (ii) and (iii) respectively).

48. Consider the following combination of atomic orbitals

Combinding orbitals (Internuclear axis)	Combinding orbitals (Internuclear axis)
(i) $s + p_x(x)$	(ii) $d_{xy} + d_{xy}(x)$
(iii) $d_{yz} + p_z(z)$	(iv) $s + s(z)$
(v) $d_{yz} + d_{yz}(x)$	(vi) $p_y + p_y(y)$
(vii) $d_{z^2} + d_{z^2}(z)$	(viii) $d_{xy} + d_{xy}(z)$
(ix) $p_x + p_x(y)$	(x) $s + p_z(x)$

Then calculate value of $a^2 + b^2 + 2cd$. (where a = σ M. O.; b = π M.O.; c = δ M.O.; d = non-bonding M.O.)

49. Consider the following table

Compounds (X are monovalent surrounding atoms)		Central atoms (A to D) belong to group	Characteristics of compounds	Number of lone pair(s) at central atom
(i)	AX_{n_1}	16	Planar and polar	m_1
(ii)	BX_{n_2}	15	Trigonal pyramidal	m_2
(iii)	CX_{n_3}	14	Zero dipole moment	m_3
(iv)	DX_{n_4}	13	All X — D — X bond angle are 120°	m_4

Then calculate value of expression $\left|\frac{n_1+n_2+n_3+n_4}{m_1+m_2+m_3+m_4}\right|^2$

50. Consider the following oxy anions

$PO_4^{3-}, P_2O_6^{4-}, SO_4^{2-}, MnO_4^-, CrO_4^{2-}, S_2O_5^{2-}, S_2O_7^{2-}$

and find the value of R + Q – P

Where

P = Number of oxy anions having three equivalent

X — O bond per central atom

Q = Number of oxy anions having two equivalent

X — O bonds per central atom

R = Number of oxy anions having four equivalent

X — O bonds per central atom

51. Consider the following complex : $[Co(NH_3)_5CO_3]ClO_4$

The coordination number, oxidation number, no. of d-electrons and number of unpaired d-electrons on the metal are respectively:

(A) 6, 2, 7, 3 (B) 7, 2, 7, 1

(C) 5, 3, 6, 4 (D) 6, 3, 6, 0

52. In the isoelectronic series of metal carbonyl, the CO bond strength is expected to increase in the order

(A) $[Mn(CO)_6]^+ < [Cr(CO)_6] < [V(CO)_6]^-$

(B) $[V(CO)_6]^- [Cr(CO)_6] < [Mn(CO)_6]^+$

(C) $[V(CO)_6]^- < [Mn(CO)_6]^+ < [Cr(CO)_6]$

(D) $[Cr(CO)_6] < [Mn(CO)_6]^+ < [V(CO)_6]^-$

53. Complexes formed in the following methods are

(i) Mond's process for purification of nickel

(ii) Removal of unreacted AgBr from photographic plate

(iii) Removal of lead poisoning from the body

	I	II	III
(A)	$Ni(CO)_4$	$[Ag(CN)_2]^-$	$[Pb\ (EDTA)]^{2-}$
(B)	$Ni(CO)_4$	$[Ag(S_2O_3)_2]^{3-}$	$[Pb\ (EDTA)]^{2-}$
(C)	$Ni(CO)_6$	$[Ag(S_2O_3)_2]^{3-}$	$[Pb\ (EDTA)]^{4-}$
(D)	$Ni(CO)_6$	$[Ag(S_2O_3)]^-$	$[Pb\ (EDTA)]^{2-}$

54. Which of the following order of CFSE is incorrect?

(A) $[Co(en)_3]^{3+} > [Co(NH_3)_6]^{3+} > [Co(H_2O)_6]^{3+}$

(B) $[PtCl_4]^{2-} > [PdCl_4]^{2-} > [NiCl_4]^{2-}$

(C) $[Ni(DMG)_2] < [Ni(en)_2]^{2+}$

(D) $[Co(ox)_3]^{3-} < [Co(en)_3]^{3+}$

55. The IR stretching frequencies of free CO, and CO in $[V(CO)_6]^-$, $[Cr(CO)_6]^-$ and $[Mn(CO)_6]^-$ are 2143 cm^{-1}, 1860 cm^{-1}, 2000 cm^{-1} and 2090 cm^{-1}, respectively. Then correct statement about metal carbonyls is

(A) C — O bond is strongest in the cation and weakest in the anion

(B) C — O bond is weakest in the cation and strongest in the anion

(C) C — O bond is longer in the cation than in the anion

(D) M — C pi bonding is higher in the cation

56. Complex compound(s) having even number of space (stereo) isomers is/are:

(where AA-symmetrical bidentate ligand and a, b, c, d, e-monodentate ligands)

(A) $[M(AA)_2b_2]^{n\pm}$ (B) $[Ma_3b_3]^{n\pm}$

(C) $[Ma_3bcd]^{n\pm}$ (D) $[Ma_2bcde]^{n\pm}$

57. Which of the following statement is true?

(A) In $[PtCl_2(NH_3)_2]^{2+}$ the cis form is optically inactive while trans form is optically active

(B) In $[Fe(C_2O_4)_3]^{3-}$, geometrical isomerism does not exist while optical isomerism exists

(C) In Mabcd, square planar complexes show both optical as well as geometrical isomerism

(D) In Mabcd tetrahedral complex, optical isomerism cannot be observed

58. Choose the correct code regarding, possible number of geometrical isomers exhibited by following complexes

(i) $[CrCl_2(NO_2)_2(NH_3)_2]^-$

(ii) $[Co(NO_2)_3(NH_3)_3]$

(iii) $[PtCl(NO_2)(NH_3)(py)]$

(iv) [PtBrCl(en)]

	I	II	III	IV
(A)	4	2	3	1
(B)	5	2	4	0
(C)	3	2	0	1
(D)	5	2	3	1

59. Consider the following two reactions

$$Cd^{2+}(aq.) + 4CH_3NH_2 \xrightarrow{K_1} A,\ \Delta G_1^\circ$$

$$Cd^{2+}(aq.) + 2H_2NCH_2CH_2NH_2 \xrightarrow{K_2} B,\ \Delta G_2^\circ$$

According to given information the correct statement(s) is/are

(A) ΔG_2° is more negative than ΔG_1°

(B) Compound A is optically inactive

(C) Compound 'B' is optically active

(D) Formation constant K_2 is greater than formation constant K_1

60. Consider the following two carbonyl compounds

(i) $[Tc(CO)_6]^+$ and (ii) $[Nb(CO)_6]^-$

Select incorrect statement(s) for given carbonyl compounds.

(A) $[Tc(CO)_6]^+$ acts as reducing agent and $[Nb(CO)_6]^-$ acts as oxidizing agent

(B) $[Nb(CO)_6]^-$ acts as reducing agent and $[Tc(CO)_6]^+$ acts as oxidizing agent

(C) Nb — C bond order in $[Nb(CO)_6]^-$ is greater than Tc — C bond order in $[Tc(CO)_6]^+$

(D) CO bond order is greater in $[Nb(CO)_6]^-$ than in $[Tc(CO)_6]$+

61. Consider the following reactions of complex compounds A, B and C

(i) $\underset{\text{(Compound A)}}{CoCl_2Br \cdot 5NH_3} + \text{excess } Ag^+(aq.) \longrightarrow 1AgCl(s)$

(ii) $\underset{\text{(Compound B)}}{CoCl_2Br \cdot 5NH_3} + \text{excess } Ag^+(aq.) \longrightarrow 2AgCl(s)$

(iii) $\underset{\text{(Compound C)}}{CoCl_2Br \cdot 4NH_3} + \text{excess } Ag^+(aq.) \longrightarrow 1AgCl(s)$

Then according to the given information the correct statement(s) is/are

(A) Compounds (A) and (B) are ionisation isomers

(B) Molar conductivity of compounds (A) and (B) are almost same

(C) Compounds (A), (B) and (C) do not exhibit geometrical isomerism

(D) Order of CFSE values : $\Delta_0(A) > \Delta_0(B) > \Delta_0(C)$

62. Select correct statement(s) regarding given complexes.

(A) $[Fe(CO)_5]$, the orbitals used for hybridization in Fe atom are s, p_x, p_y, p_z, d_{z^2} and it is high spin complex

(B) $[Pt(NH_3)_2Cl_2]$, the orbitals used for hybridization on Pt atom are s, p_x, p_y, $d_{x^2-y^2}$ and it is low spin complex

(C) $[Cr(H_2O)_6]^{3+}$, the orbitals used for hybridization in Cr atom are s, p_x, p_y, p_z, $d_{z^2}, d_{x^2-y^2}$ and it is high spin complex

(D) $Ni(CO)_4$, the orbitals used for hybridization in Ni atom are, s, p_x, p_y, p_z, and it is low spin complex

63. A d-block element forms octahedral complex but its magnetic moment remains same either in strong field or in weak field ligand. Which of the following is/are correct?

(A) Element always forms colourless compound

(B) Number of electrons in t_{2g} orbitals are higher than in e_g, orbitals

(C) It can have either d^3 or d^8 configuration

(D) It can have either d^7 or d^8 configuration

64. Consider the following complex compounds

(i) $[Cu(NH_3)_4)\ [Cu(NO_2)_4]$

(ii) $[Cr(py)_6)\ [Cr(SCN)_6]$

(iii) $[Co(NH_3)_5\ (NO_2)\ ]\ [Pt(SCN)_4]$

If x_1, x_2, x_3 are more possible coordination isomers of given complex compounds respectively, then calculate the value of $x_1^2 + x_2^2 + x_3^2$.

65. Consider the following ligands NH_2^-, $acac^-$, OH^-, gly^-, O_2^-, o-Phen, $Hdmg^-$, NO_2^-, CO_3^{2-}, Cl^-, CH_3COO^-, en, SO_4^{2-}. Then calculate the value of "P + Q – R – S".

Here

P : Total numbers of ligands which act as bridging as well as monodentate only

Q : Total number of flexidentate ligands

R : Total number of bidentate ligands only

S : Total number of unsymmetrical bidentate ligands

66. Consider the following reactions at 1000°C

(i) $Zn(s) + \frac{1}{2}O_2(g) \xrightarrow{\Delta} ZnO(g);\ \Delta G° = -360\ kJ\ mol^{-1}$

(ii) $C(s) + \frac{1}{2}O_2(g) \xrightarrow{\Delta} CO(g);\ \Delta G° = -460\ kJ\ mol^{-1}$

and choose the correct statement at 1000°C.

(A) ZnO is more stable than CO

(B) ZnO can be reduced to Zn by C

(C) ZnO and CO are formed at equal rate

(D) ZnO can not be reduced to Zn by C

67. The metal X is prepared by the electrolysis of fused chloride. It reacts with hydrogen to form a colourless solid from which hydrogen gas is released on treatment with water. The metal is

(A) Al (B) Ca

(C) Cu (D) Zn

68. $XCl_2(\text{excess}) + YCl_2 \longrightarrow XCl_4 + Y\downarrow$

$YO \xrightarrow[>400°]{\Delta} \frac{1}{2}O_2 + Y$, Ore of Y would be

(A) Siderite

(B) Cinnabar

(C) Malachite

(D) Horn silver

69. Consider the following metallurgical processes.

(i) Heating impure metal with CO and distilling the resulting volatile carbonyl (b.p. 43°C) and finally decomposition at 150°–200°C to get the pure metal

(ii) Heating the sulphide ore in air until a part is converted to oxide and then further heating in the absence of air to let the oxide react with unchanged metal sulphide

(iii) Electrolysis of the molten electrolyte containing approximately equal amounts of the metal chloride and NaCl to obtained the metal

"The processes used for obtaining magnesium, nickel and copper are respectively :

(A) (i), (ii) and (iii)

(B) (ii), (iii) and (i)

(C) (iii), (i) and (ii)

(D) (ii), (i) and (iii)

70. In the extraction of aluminium

Process X: applied for red bauxite to remove iron oxide (chief impurity)

Process Y: (Serpeck's process) : applied for white bauxite to remove Z (chief impurity) then, process X and impurity Z are :

(A) X = Hall and Heroult's process and Y = SiO_2

(B) X = Bayer's process and Y = SiO_2

(C) X = Serpeck's process and Y = iron oxide

(D) X = Bayer's process and Y = iron oxide

71. $FeCr_2O_4$ (chromite) is converted to Cr by following steps

$$\text{Chromite} \xrightarrow{i} Na_2CrO_4 \xrightarrow{ii} Cr_2O_3 \xrightarrow{iii} Cr$$

Reagents in i, ii and iii step might be

	i-Step	**ii-Step**	**iii-Step**
(A)	Na_2CO_3/air, Δ	C	C
(B)	NaOH/air, Δ	C, Δ	Al, Δ
(C)	Na_2CO_3/air, Δ	C, Δ	C, Δ
(D)	Conc. H_2SO_4, Δ	NH_4Cl, Δ	C, Δ

72. Blister copper is refined by stirring molten impure metal with green logs of wood because such a wood liberates hydrocarbon gases (like CH_4). This process X is called ..... and the metal contains impurities of Y is .....

(A) X = cupellation, Y = CuO_2

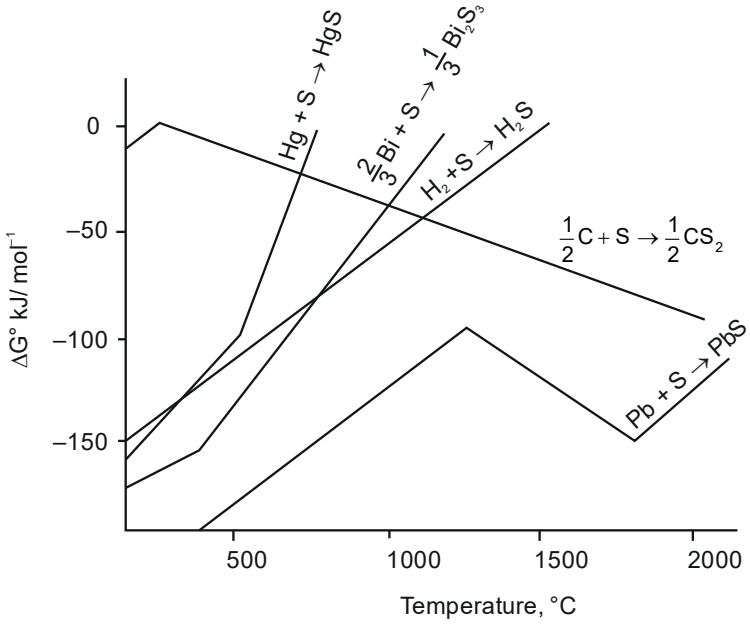
(B) X = polling, Y = Cu_2O

(C) X = polling, Y = CuO

(D) X = cupellation, Y = CuO

Paragraph for Q. Nos. 73 to 75

The Ellingham diagram for a number of metallic sulphides is shown below.



73. Formation of which of the sulphides is most spontaneous?

(A) HgS (B) Bi_2S_3

(C) PbS (D) CS_2

74. Which sulphide occurs to minimum extent in nature?

(A) HgS (B) H_2S

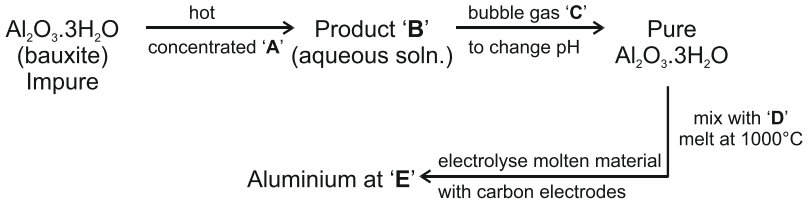
(C) Bi_2S_3 (D) CS_2

75. Which of the following sulphides can not be reduced to metal by H_2 at about 1000°C?

(A) HgS (B) PbS

(C) Bi_2S_3 (D) All of these

76. In the extraction of aluminium metal, one of the process is summarised as follows



Which of the following entries correctly summarises reagents, electrodes and products of the process?

	A	**B**	**C**	**D**	**E**
(A)	NaOH	Al^{3+}	HF	Na_3AlF_6	Cathode
(B)	NaOH	$NaAlO_2$	CO_2	NaF	Anode
(C)	H_2SO_4	$Al_2(SO_4)_3$	NH_3	Na_3AlF_6	Cathode
(D)	NaOH	$NaAlO_2$	CO_2	Na_3AlF_6	Cathode

77. Column-I and Column-II contains four entries each. Entries of Column-I are to be matched with some entries of Column-II. Each entry of column-I may have the matching with one or more than one entries of Column-II.

Column-I (Main ore of metal)	**Column-II (Process involved in commercial extraction pure metal)**
(A) Cinnabar	(P) Froth floatation method
(B) Chalcopyrite	(Q) Roasting
(C) Bauxite	(R) Distillation
(D) Argentite	(S) Leaching
	(T) Calcination

78. When a concentrated solution of ammonia is saturated with sodium chloride in the presence of pieces of dry ice, a water cloud forms. This is due to the

(A) Precipitation of sodium carbonate from the reaction mixture

(B) Precipitation of sodium hydrogen carbonate from the reaction mixture

(C) Precipitation of ammonium hydrogen carbonate from the mixture

(D) Precipitation of ammonium carbonate

79. A compound (A) is used in preparation of washing soda to recover ammonia in Solvay's process. When CO_2 is bubbled through an aqueous solution of (A), the solution turns milky. It is used in white washing due to disinfectant nature. What is the chemical formula of A?

(A) $Ca(HCO_3)_2$ (B) CaO

(C) $Ca(OH)_2$ (D) $CaCO_3$

80. $KO_2 + CO_2 + H_2O \xrightarrow{\text{more } CO_2} [X] + [Y]$

Products [X] and [Y] are respectively

(A) K_2CO_3, O_2 (B) $KHCO_3$, O_2

(C) KOH, K_2CO_3 (D) $KHCO_3$, H_2O

81. Baking powder used to make cake is a mixture of starch, $NaHCO_3$ and $Ca(H_2PO_4)_2$. The function of $Ca(H_2PO_4)_2$ is

(A) To show down the release of CO_2 gas

(B) Being acidic in nature and gives CO_2 when moistened with $NaHCO_3$

(C) To act as a filler

(D) None of these

82. $A + H_2O \longrightarrow NaOH$; $A \xrightarrow[400°C]{O_2} B \xrightarrow[\text{at } 25°C]{H_2O} NaOH + O_2$

B is used for oxygenating in submarine. A and B are

(A) Na_2O_2 and Na_2O (B) Na_2O and Na_2O_2

(C) Na_2O_2 and O_2 (D) Na_2O and O_2

83. Calcium imide on hydrolysis gives gas (B) which on oxidation by bleaching powder gives gas (C). Gas (C) on reaction with magnesium give compound (D) which on hydrolysis gives again gas (B). Identify (B), (C) and (D).

(A) NH_3, N_2, Mg_3N_2

(B) N_2, NH_3, MgNH

(C) N_2, N_2O_5, $Mg(NO_3)_2$

(D) NH_3, NO_2, $Mg(NO_2)_2$

84. The more commonly used baking powder contains about 30% $NaHCO_3$, 20% $NaAl(SO_4)_2$, 10% $Ca(H_2PO_4)_2$ and 40% starch. Which of the following statements is/are correct?

(A) $Ca(H_2PO_4)_2$ is acidic and when moistened it reacts with $NaHCO_3$ evolving CO_2 gas

(B) $NaAl(SO_4)_2$ slows down the decomposition reaction of $NaHCO_3$ so that CO_2 is evolved more slowly

(C) $NaAl(SO_4)_2$ is acidic and when moistened it reacts with $NaHCO_3$ evolving CO_2

(D) Both (A) and (B)

Paragraph for Q. No. 85

Most metal oxides are thermally stable at temperatures upto 1000°C but the oxide of metals below hydrogen in the electrochemical series decompose fairly easily. Thus HgO and Ag_2O decompose on heating.

85. Correct code for following thermal decomposition reaction(s) evolving gas having equal number of σ- and π- bonds is

(i) $BeCO_3 \xrightarrow{\Delta}$

(ii) $ZnSO_4 \xrightarrow[>800°C]{\Delta}$

(iii) $FeSO_4 \xrightarrow{<300°C}$

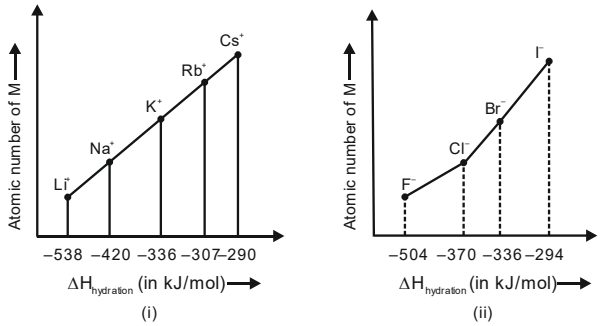
(iv) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$

(A) i, ii (B) i, iii, iv

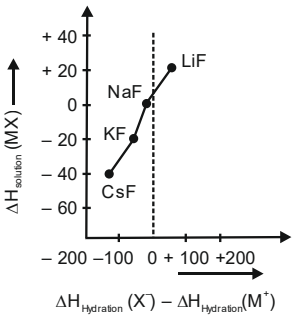
(C) i, ii, iii (D) All of these

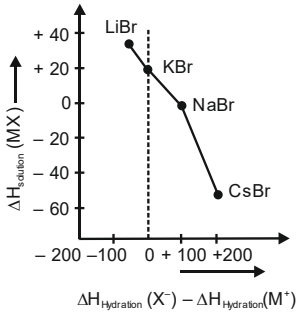
86. Consider the following two graphs between atomic number of element and hydration enthalpies (at 25°C) of corresponding ion.

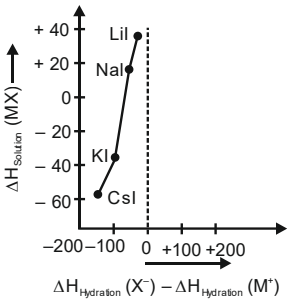
(M = Alkali metal and X = halogens)



Then according to the given information which of the following is/are correct relationship between enthalpy of solution ($\Delta H_{solution}$) of a salt MX in water and difference of the enthalpies of hydration ($\Delta H_{hydration}$) of the constituent ion (M^+ and X^-)?

(A) 

(B) 

(C) 

(D) None of these

87. The silicate anion in the mineral kinoite is a chain of three SiO_4 tetrahedra, that share corners with adjacent tetrahedra. The charge of the silicate anion is:

(A) –4 (B) –8

(C) –6 (D) –2

88. $H_2C_2O_4 \xrightarrow{\Delta} \text{gas (A)} + \text{gas (B)} + \text{liquid (C)}$

Gas (A) burns with a blue flame and is oxidised to gas (B)

$\text{Gas (A)} + Cl_2 \longrightarrow (D) \xrightarrow{NH_3, \Delta} (E)$

A, B, C and E are

(A) CO_2 , CO, H_2O, $HCONH_2$

(B) CO, CO_2, $COCl_2$, $HCONH_2$

(C) CO, CO_2, H_2O, NH_2CONH_2

(D) CO, CO_2, H_2O, $COCl_2$

89. $BX_3 + NH_3 \xrightarrow{\text{R.T.}} BX_3 \cdot NH_3 + \text{Heat of adduct formation } (\Delta H)$

The numerical value of ΔH is found ro be maximum for

(A) BF_3 (B) BCl_3

(C) BBr_3 (D) BI_3

90. Choose the correct sequence for the geometry of the given molecules.

$Al_2(CH_3)_6$, Borazole, $B_3O_6^{3-}$, Fe_2Cl_6, I_2Cl_6

['P' stands for planar and 'NP' stands for non-planar]

(A) NP, NP, NP, P, P (B) P, P, NP, NP, P

(C) NP, NP, NP, P, NP (D) NP, P, P, NP, P

91. The correct code for stability, of oxidation states for given cations is

(i) $Pb^{2+} > Pb^{4+}$, $Tl^{+} < Tl^{3+}$

(ii) $Bi^{3+} > Sb^{3+}$, $Sn^{2+} < Sn^{4+}$

(iii) $Pb^{2+} > Pb^{4+}$, $Bi^{3+} > Bi^{5+}$

(iv) $Tl^{3+} < In^{3+}$, $Sn^{2+} > Sn^{4+}$

(v) $Sn^{2+} < Pb^{2+}$, $Sn^{4+} > Pb^{4+}$

(vi) $Sn^{2+} < Pb^{2+}$, $Sn^{4+} < Pb^{4+}$

(A) (v) and (vi) (B) (i), (iii) and (vi)

(C) (iii) and (v) (D) (ii) and (iv)

92. Two oxides of nitrogen, NO and NO_2 are allowed to react together at 243 K and form a coloured compound of nitrogen (X). When compound (X) reacts with water to yield another compound of nitrogen (Y). The geometry of the anion of (Y) molecule is

(A) Triangular planar (B) Triangular pyramidal

(C) Tetrahedron (D) Square planar

93. $NH_3 + O_2 \xrightarrow[\Delta]{Pt} A + H_2O$;

$A + O_2 \longrightarrow B$;

$B + O_2 + H_2O \longrightarrow C$

A, B and C are

(A) N_2O, NO_2 and HNO_3

(B) NO, NO_2 and HNO_3

(C) NO_2, NO and HNO_3

(D) N_2O, NO and HNO_3

94. Which of the following properties of halogens increase with increasing atomic number?

(i) Ionization energy

(ii) Ionic radius

(iii) Bond energy of the X_2 molecule

(iv) Enthalpy of vaporisation

(A) (i), (ii), (iii) (B) (i), (iii)

(C) (ii), (iv) (D) (iv)

95. $Cl_2(g) + Ba(OH)_2 \longrightarrow X(aq.) + BaCl_2 + H_2O$

$X + H_2SO_4 \longrightarrow Y + BaSO_4$

$Y \xrightarrow[\Delta > 365\,K]{\Delta} Z + H_2O + O_2$

Y and Z are respectively

(A) $HClO_4$, ClO_2 (B) $HClO_3$, ClO_2

(C) $HClO_3$, ClO_6 (D) $HClO_4$, Cl_2O_7

96. The three elements X, Y and Z with electronic configurations shown below all form hydrides

Element	Electronic configuration
X	$1s^2 2s^2 2p^2$
Y	$1s^2 2s^2 2p^6 3s^1$
Z	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^5$

Which set of properties match correctly with properties of the hydrides of these elements

	Hydride of X	Hydride of Y	Hydride of Z
(A)	Colourless gas insoluble in H_2O	Silver/grey solid reacts with H_2O to form an alkali	Colourless gas forms a strong acid in H_2O
(B)	Colourless liquid, no reaction with H_2O	Silver/grey solid forms H_2 and H_2O	Ionic solid with formula ZH
(C)	Non-polar compound reacts with Cl_2 in light	Silver/grey ionic solid with formula YH_2	Forms when water is added to phosphorus and elemental Z
(D)	Colourless gas which burns with air	Silver/grey solid which reacts violently with acids	Colourless, corrosive liquid at STP

97. Consider the following transformations

(i) $XeF_6 + NaF \longrightarrow Na^+ [XeF_7]^-$

(ii) $2PCl_5(s) \longrightarrow [PCl_4]^+ [PCl_6]^-$

(iii) $[Al(H_2O)_6]^{3+} + H_2O \longrightarrow [Al(H_2O)_5OH]^{2+} + H_3O^+$

Possible transformations are

(A) (i), (ii), (iii) (B) (i), (iii)

(C) (i), (ii) (D) (ii), (iii)

98. Among the following, cyclic species are

(i) $H_5P_3O_{10}$

(ii) $[B_3O_3(OH)_5]^{2-}$

(iii) Trimer of $SO_3(S_3O_9)$

(iv) $P_3N_3Cl_6$

(A) (i), (iii) (B) (i), (iii), (iv)

(C) (ii), (iii), (iv) (D) (i), (ii), (iii), (iv)

Paragraph for Q. Nos. 99 to 102

(i) $P + C \text{ (carbon)} + Cl_2 \longrightarrow Q + CO\uparrow$

(ii) $Q + H_2O \longrightarrow R + HCl$

(iii) $BN + H_2O \longrightarrow R + NH_3\uparrow$

(iv) $Q + LiAlH_4 \longrightarrow S + LiCl + AlCl_3$

(v) $S + H_2O \longrightarrow R + H_2\uparrow$

(vi) $S + NaH \longrightarrow T$

(P, Q, R, S and T do not represent their chemical symbols)

99. Compound Q has

(i) Zero dipole moment
(ii) A planar trigonal structure
(iii) An electron deficient compound
(iv) A Lewis base

Choose the correct code

(A) (i), (iv)
(B) (i), (ii), (iv)
(C) (i), (ii) (iii)
(D) (i), (ii), (iii), (iv)

100. Compound T is used as a/an

(A) Oxidising agent
(B) Complexing agent
(C) Bleaching agent
(D) Reducing agent

101. Compound S is

(i) An oldd-e^- compound
(ii) (2c – $3e^-$) compound
(iii) A electron deficient compound
(iv) A sp^2 hybridized compound

Choose the correct code

(A) (iii)
(B) (i), (iii)
(C) (ii), (iii), (iv)
(D) (i), (ii), (iv)

102. Consider the following reactions

$$A_x + yB_2 \xrightarrow[\text{supply of air}]{\text{limited}} \text{compound P} \xrightarrow[\text{excess air}]{+zB_2} \text{Compound Q}$$

If atomic number of elements A and b are 15 and 8 respectively, then according to the given information the correct statement(s) is/are

(A) (B — A — B) bond angle in compound Q > (B — A — B) bond angle in compound P
(B) (A — B) bond length in compound Q < (A — B) bond length in compound P
(C) Value of x + y + z is 9
(D) Compound P is P_2O_5 and compound Q is P_4O_{10}

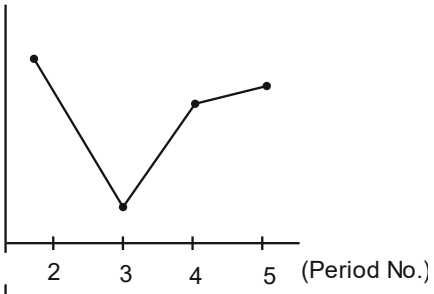
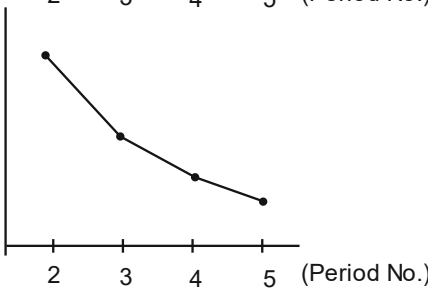
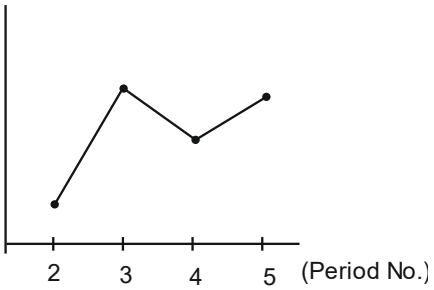
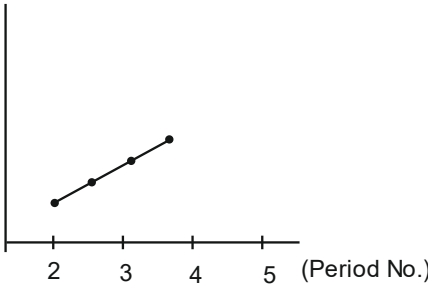
103. Consider the following table

Compound	Enthalpy of formation (kJ mol^{-1})	Bond angle (H — C.A. — H)	Boiling point (°C)
H_2O	x_1	y_1	z_1
H_2S	x_2	y_2	z_2
H_2Se	x_3	y_3	z_3
H_2Te	x_4	y_4	z_4

According to given information the correct order is/are

(A) $x_1 > x_2 > x_3 > x_4$
(B) $y_1 > y_2 > y_3 > y_4$
(C) $z_1 > z_4 > z_3 > z_2$
(D) $x_1 > x_4 > x_3 > x_2$

104. Match the Column-I with the graph of Column-II which is most close to the answer.

Column-I	Column-II
(A) Bond energies of the hydra acids of the halogens fluorine, chlorine, bromine, iodine	(P) 
(B) Boiling points of the hydrides of the 16 group elements oxygen, sulphur, selenium, tellurium	(Q) 
(C) The stability of monochlorides of group 13 elements boron, aluminium, gallium, indium	(R) 
(D) Melting points of the dioxides of the group 14 elements carbon, silicon, germanium, tin	(S) 

105. Consider the structure of Al_2Me_6 compound and find the value of $\frac{x-y}{z}$

Where x = Maximum number of atoms that can lie in plane having terminal (Al – Me) bonds.

y = Total number of 3c – $2e^-$ bonds

z = Total number of atoms that are sp^3 hybridized

106. Consider the following covalent compounds in their solid state and find the value of expression (X + Y + Z)

N_2O_5, Cl_2O_6, PCl_5, I_2Cl_6, XeF_6, PBr_5

Where X = Total number of compounds in which central atom of cationic or anionic part is sp^3 hybridized

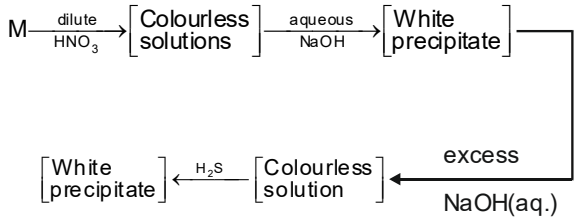
Y = Total number of compounds having 90° bond angle either in cationic or anionic part

Z = Total number of compounds having 109°28' bond angle either in cationic or anionic part

107. When B_2H_6 is allowed to react with following Lewis bases, then how many given Lewis bases form adduct through symmetrical cleavage of B_2H_6?

NH_3, $MeNH_2$, Pyridine, CO, T.H.F, PH_3, PF_3, Me_3N, Me_2NH

108. A metal M and its compound can give the following observable changes in a consequence of reactions.



(A) Mg (B) Pb

(C) Zn (D) Sn

109. $AgNO_3 \xrightarrow{\Delta} (W) + (X) + O_2$

(X) + $H_2O \longrightarrow HNO_2 + HNO_3$

(W) + $HNO_3 \longrightarrow Y + NO + H_2O$

(Y) + $Na_2S_2O_3$ (excess) $\longrightarrow$ (Z) + $NaNO_3$

Identify (W) to (Z)

(A) W = Ag X = N_2O Y = $AgNO_3$ Z = $Na_2[Ag(S_2O_3)_2]$

(B) W = Ag_2O X = NO Y = $AgNO_3$ Z = $Na_3[Ag(S_2O_3)_2]$

(C) W = Ag X = NO_2 Y = $AgNO_3$ Z = $Na_3[Ag(S_2O_3)_2]$

(D) W = Ag_2O X = N_2 Y = $AgNO_3$ Z = $Na[Ag(S_2O_3)_2]$

110. Limestone is present in the blast furnace production of iron in order to

(i) Provide a source of CaO

(ii) Remove some impurities

(iii) Supply CO_2

(A) (i), (ii), (iii) (B) (i), (ii)

(C) (ii), (iii) (D) (i) only

111. When aqueous sodium hydroxide is added to an aqueous solution of chromium (III) ions, a green blue precipitate is first formed which re-dissolves to give a green solution. This green colour is due to

(A) $[Cr(H_2O)_6]^{3+}$ (B) CrO_4^{2+}

(C) $[Cr(OH)_4]^-$ (D) $[Cr(OH)_3(H_2O)_3]$

112. $FeCr_2O_4 + Na_2CO_3 + O_2 \xrightarrow{\text{Fusion}} [X] \xrightarrow[H_2O]{H^+} [Y] \xrightarrow{H_2O_2/H^+} [Z]$

Which of the following statement is true for the compounds [X], [Y] and [Z]?

(A) In all three compounds, the chromium is in +6 oxidation state

(B) [Z] is a deep blue-violet coloured compound which decomoposes rapidly in aqueous solution into Cr^{+3} and dioxygen

(C) Saturated solution of [Y] gives bright orange compound, chromic anhydride, with concentrated H_2SO_4

(D) All of these

113. Consider the following transformation

$$2CuX_2 \xrightarrow{\text{Room temperature}} 2CuX + X_2 \uparrow$$

Then X^- can be

(A) F^-, Br^- (B) Cl^-, Br^-

(C) CN^-, I^- (D) Cl^-, F^-

Paragraph for Q. No. 114

Iron (+II) is one of the most important oxidation states, and salts are called ferrous salts. Most of the Fe (+II) salts are pale green and contain $[Fe(H_2O)_6]^{2+}$ ion. Fe(+II) compounds are easily oxidised by air and so are difficult to obtain pure Fe^{2+} from many complexes like $K_4[Fe(CN)_6]$

114. $FeSO_4$ is used in brown ring test for nitrates and nitrites. In this test, a freshly prepared $FeSO_4$ solution is mixed with solution containing NO^-_2 or NO_3^- and the conc. H_2SO_4 is run down the side of the test tube. If the mixture gets hot or is shaken.

(i) The brown colour disappear

(ii) NO is evolved

(iii) A yellow solution of $Fe_2(SO_4)_3$ is formed

(A) (i), (ii), (iii) correct (B) (i), (iii) correct

(C) (ii), (iii) correct (D) Only (i) correct

115. Consider the following sequence of reactions

$$M^{2+}(aq.) \xrightarrow{NH_4Cl(s)+(NH_4)_2CO_3\,Sol.} Q \downarrow \xrightarrow[\text{Followed by addition of } (NH_4)_2C_2O_4]{CH_3COOH} R \downarrow$$

Which of the following cation can form ppt. Q but does not form ppt. R?

(A) Mg^{2+}(aq.) (B) Ca^{2+}(aq.)

(C) Sr^{2+}(aq.) (D) Ba^{2+}(aq.)

116. $M(\text{Salt}) + \text{Dil. HCl} \xrightarrow{\text{Warm}} N \uparrow + P \downarrow$

gas N changes colour of $FeSO_4$ solution into yellow solution then salt M in above reaction is

(A) BaS_2O_3 (B) Ag_2SO_3

(C) $AgNO_2$ (D) $Pb(NO_3)_2$

117. Sodium salt solution + $AgNO_3$ soln. $\longrightarrow$ Coloured precipitate.

If coloured precipitate is soluble in both dil. HNO_3 and excess conc. NH_3 solution then which of the following anion is present in the salt solution?

(A) S^{2-}(aq.) (B) I^-(aq.)

(C) $PO_4^{3-}(aq,)$ (D) Br^-(aq.)

118. Fe(Finely powdered) + HCl(dil.) $\longrightarrow$ P + Q $\uparrow$

compound P does not precipitate with

(A) $AgNO_3$ (B) $K_3[Fe(CN)_6]$

(C) KCN (excess) (D) $NH_4Cl + NH_4OH$

119. Bromine is commercially prepared from sea water by displacement reaction

$Cl_2 + 2Br^-(aq.) \longrightarrow 2Cl^-(aq.) + Br_2$

Br_2 gas thus formed is dissolved into solution of Na_2CO_3 and then pure Br_2 is obtained by treatment of the solution with

(A) $Ca(OH)_2$ (B) NaOH

(C) H_2SO_4 (D) HI

120. In a closed container there is a mixture of SO_2, CO_2 and O_2 gas. Which sequence of reagent can be helpful to separate them?

(i) Limewater

(ii) Acidified potassium dichromate

(iii) Alkaline pyrogallol

(A) (i), (ii) and (iii) (B) (ii), (i), (iii)

(C) (iii), (ii), (i) (D) (iii), (i), (ii)

121. $Xe + F_2 \xrightarrow{1:20} X \xrightarrow{H_2O} Y \xrightarrow{H_2O} Z \xrightarrow{H_2O} XeO_3$

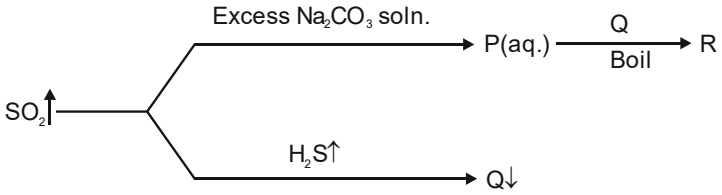
Select correct option(s) for X, Y, Z and given chemical change

(A) X, Y and Z are in same oxidation state

(B) All have equal number of lone pair on central atom

(C) All are non-planar

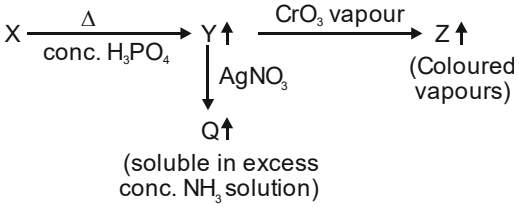
(D) All have equal number of covalent bonds

122. 

Incorrect statement about R is

(A) Antichlor agent (B) Fixing agent in photography

(C) Forms ppt. with $CaCl_2$ solution (D) Reduces $Cu^{2+}(aq,)$ cation

123. 

Which of the following anion cannot be in X?

(A) F^- (B) Cl^-

(C) Br^- (D) I^-

124. Find the value of expression |x – y| for following compounds.

where,

x = Total number of water insoluble salts.

y = Total number of salts, which can liberate non-polar acidic gas during their complete thermal decomposition,

$BaCO_3$, $PbSO_4$, $AgNO_3$, CaC_2O_4, $CsHCO_3$,

Na_3PO_4, CH_3COOAg, $Mg(OH)_2$, $Pb(NO_3)_2$

125. A reddish pink substance on heating gives off a vapour which condenses on the sides of the test tube and the substance turns blue. It on cooling water is added to the residue it turns to its original colour. The substance is :

(A) Iodine crystals (B) Copper sulphate crystals

(C) Cobalt chloride crystals (D) Zinc oxide

126. A colourless water soluble solid 'X' on heating gives equimolar quantities of Y and Z. Y gives dense white fumes HCl and Z does so with NH_3. Y gives brown precipitate with Nessler's reagent and Z gives white precipitate with nitrates of Ag^+, Pb^{2+} and Hg^+. 'X' is:

(A) NH_4Cl (B) NH_4NO_3

(C) NH_4NO_2 (D) $FeSO_4$

127. A white crystalline salt imparts a violet colour to Bunsen flame, and with hot concentrated H_2SO_4, forms a pungent gas. On treatment with an $AgNO_3$ solution, this gas forms a white precipitate readily soluble in NH_3. The white crystalline salt may be

(A) Na_2SO_4 (B) KCl

(C) $CaCl_2$ (D) $SrCl_2$

128. A white solid gives a green residue on being subjected to the cobalt nitrate test. On being warmed with concentrated H_2SO_4, the solid gives a brown gas, which evolves vigorously on the addition of Cu turnings. The solid may be

(A) $Zn(NO_3)_2$ (B) $Al(NO_3)_2$

(C) $ZnBr_2$ (D) $Mg(NO_3)_2$

129. A compound 'X' on heating gives a colourless gas. The residue is dissolved in water to obtain 'Y'. Excess CO_2 is passed through aqueous solution of 'Y' when 'Z' is formed. 'Z' on gentle heating gives back 'X'. The compound 'X' is:

(A) $NaHCO_3$ (B) Na_2CO_3

(C) $Ca(HCO_3)_2$ (D) $CaCO_3$

130. Solid KCl, when heated with solid $K_2Cr_2O_7$ and concentrated H_2SO_4, gives red vapours (a) that turn NaOH solution yellow (b). The yellow solution, when acidified with acetic acid and treated with lead acetate, gives a yellow precipitate (C). Which of the following is true with respect to a, b and c?

(A) a and b contain CrO_4^{2-}

(B) a and c contain CrO_4^{2-}

(C) a, b and c contain CrO_4^{2-}

(D) a, b and c contain Cr(VI)

131. Choose the correct code by identifying (X), (Y) and (Z) in each case for the changes indicated:

(i) $CrO_2Cl_2 \xrightarrow{KOH} (X) \xrightarrow{\text{conc. } H_2SO_4} (Y) \xrightarrow{AgNO_3} (Z)$

(ii) $CrCl_3(aq.) \xrightarrow[NaOH]{\text{excess}} (X) \xrightarrow[H_2O,\text{ boil}]{Na_2O_2} (Y) \xrightarrow[\text{acetate}]{\text{lead}} (Z)$

(iii) $ZnSO_4(aq.) \xrightarrow{Na_2CO_3} (X) \xrightarrow{\Delta} (Y) \xrightarrow[\text{nitrate, }\Delta]{\text{cobalt}} (Z)$

(iv) $CuCl_2(aq.) \xrightarrow[H_2S]{NH_4OH} (X) \xrightarrow[\Delta]{HNO_3} (Y) \xrightarrow[\text{excess}]{KCN} (Z)$

(A) X = K_2CrO_4 Y = $K_2Cr_2O_7$ Z = Ag_2CrO_4

(B) X = $[Cr(OH)_4]^-$ Y = Na_2CrO_4 Z = $PbCrO_4$

(C) X = $ZnCO_3$ Y = ZnO Z = $CoZnO_2$

(D) X = CuS Y = $Cu(NO_3)_2$ Z = $K_3[Cu(CN)_4]$

132. A sulphate of a metal (A) on heating evolves two gases (B) and (C) and an oxide (D). Gas (B) turns $K_2Cr_2O_7$ paper green while gas (C) forms a trimer in which there is no S-S bond. Compound (D) with conc. HCl forms a Lewis acid (E) which exists in a dimer. Compounds (A), (B), (C), (D) and (E) are respectively

(A) $FeSO_4$, SO_2 , SO_3, Fe_2O_3, $FeCl_3$

(B) $Al_2(SO_4)_3$, SO_2, SO_3, Al_2O_3, $FeCl_3$

(C) FeS, SO_2 , SO_3 , $FeSO_4$, $FeCl_3$

(D) FeS, SO_2, SO_3, $Fe_2(PO_4)_3$, $FeCl_2$

133. An organic compound (A) on heating produces two gases (B) and (C) and neutral oxide (D) which turns cobalt chloride paper pink. Gas (B) turns lime water milky and produces an acidic solution with water. Gas (C) produces a poisonous gas (E) with chlorine gas, this gas with ammonia gives an organic compound (F) which on further reaction with (D) gives NH_3 gas.

Then, compounds (A) and (F) can be found as:

(A) $H_2C_2O_4$ and NH_2CONH_2

(B) CH_3COOH and NH_2CONH_2

(C) $CHCl_3$ and CH_3CONH_2

(D) CH_3Cl and NH_2COONH_4

134. A white solid forms Rinmann's green in the charcoal cavity test in an oxidising flame. On treatment with dilute H_2SO_4, this solid produces a gas that turns an acidified dichromate paper green and lead acetate paper black. The white solid is

(A) PbS

(B) $ZnSO_3$

(C) ZnS

(D) Na_2S

135. $\underset{\text{Imparts violet colour to flame}}{(T)} \xrightarrow{\text{compound (U)} + \text{conc. } H_2SO_4} \underset{\text{Red gas}}{(V)} \xrightarrow{NaOH + AgNO_3} \underset{\text{Red ppt.}}{(W)} \xrightarrow{NH_3 \text{ soln.}} (X)$

$(W) \xrightarrow{\text{dil. HCl}} (Y)$ white ppt.

$(U) \xrightarrow[\Delta]{NaOH} (Z)$ gas (gives white fumes with HCl)

Identify (T) to (Z).

(A) T = $KMnO_4$, U = HCl, V = Cl_2, W = HgI_2, X = $Hg(NH_2)NO_3$, Y = Hg_2Cl_2, Z = N_2

(B) T = $K_2Cr_2O_7$, U = NH_4Cl, V = CrO_2Cl_2, W = Ag_2CrO_4, X = $[Ag(NH_3)_2]^+$, Y = AgCl, Z = NH_3

(C) T = K_2CrO_4, U = KCl, V = CrO_2Cl_2, W = HgI_2, X = Na_2CrO_4, Y = $BaCO_3$, Z = NH_4Cl

(D) T = K_2MnO_4, U = NaCl, V = CrO_3, W = $AgNO_2$, X = $(NH_4)_2CrO_4$, Y = $CaCO_3$, Z = SO_2

136. A white powder solid A forms a light green solution with water, which on treatment with potassium hexacyanoferrate(III) gives a blue precipitate. On being strongly heated, A leaves a brown residue and forms a mixture of two gaseous oxides, which turns a dichromate solution green and forms a white precipitate with a $BaCl_2$ solution containing concentrated HCl, 'A' is:

(A) $CuSO_4$ (B) $Fe_2(SO_4)_3$

(C) $FeSO_4$ (D) $Cr_2(SO_4)_3$

137. A white sublimable solid, when boiled with an NaOH solution, gives a colourless gas that turns Nessler's reagent brown. The solid, on being heated with solid $K_2Cr_2O_7$ and concentrated H_2SO_4, gives red brown vapours. The white solid can be

(A) NH_4I (B) NH_4Br

(C) NH_4Cl (D) $(NH_4)_2SO_4$

138. $\underset{\text{Black}}{X(s)} \xrightarrow{\text{dil. HCl}} Y \uparrow \xrightarrow[NaOH]{[Na_2(Fe(CN))_5(NO)]}$ Purple solution Gas Y has been allowed to react with following species in neutral/acidic medium

(A) $FeCl_3$ (B) $CuSO_4$

(C) $BaCl_2$ (D) SO_2

(E) $Cr_2O_7^{2-}$ (F) CH_3COONa

(G) Hg^{2+}

Then calculate value of (P + Q – R)

P : Number of species which undergoes redox reaction with gas Y

Q : Number of species which give black ppt with gas Y

P : Number of species with which gas Y produce no observable change

❑ ❑ ❑

Book Crunch

Mathematics

MATHEMATICS

1. The value of a and b for which $|e^{|x-b|} - a| = 2$, has four distinct solutions, are

   (1) $a \in (-3, \infty), b = 0$ (2) $a \in (2, \infty), b = 0$

   (3) $a \in (3, \infty), b \in R$ (4) $a \in (2, \infty), b = a$

2. Find the number of real ordered pair(s) (x, y) for which $16^{x^2+y} + 16^{x+y^2} = 1$

   (1) 0 (2) 1

   (3) 2 (4) 3

3. Number of ordered pair (a, b) from the set $A = \{1, 2, 3, 4, 5\}$ so that the function $f(x) = \frac{x^3}{3} + \frac{a}{2}x^2 + bx + 10$ is an injective mapping $\forall\ x \in R$

   (1) 13 (2) 14

   (3) 15 (4) 16

4. Number of elements in the range set of $f(x) = \left[\frac{x}{15}\right]\left[-\frac{15}{x}\right] \forall\ x \in (0, 90)$; (where [.] denotes greatest integer function)

   (1) 5 (2) 6

   (3) 7 (4) Infinite

5. Let $f(x) = \begin{bmatrix} 2x+3 & ; & x \le 1 \\ a^2x+1 & ; & x > 1 \end{bmatrix}$. If the range of $f(x) = R$ (set of real numbers) then number of integral value(s), which a may take

   (1) 2 (2) 3

   (3) 4 (4) 5

6. The function, $f(x) = \begin{cases} \dfrac{(x^{2n})}{(x^{2n}\operatorname{sgn} x)^{2n+1}}\left(\dfrac{e^{\frac{1}{x}} - e^{-\frac{1}{x}}}{e^{\frac{1}{x}} + e^{-\frac{1}{x}}}\right) & x \neq 0 \\ 1 & x = 0 \end{cases}$ $n \in N$ is

   (1) Odd function (2) Even function

   (3) Neither odd nor even function (4) Constant function

7. Let $f : R - \left\{\frac{3}{2}\right\} \to R$, $f(x) = \frac{3x+5}{2x-3}$. Let $f_1(x) = f(x)$, $f_n(x) = f(f_{n-1}(x))$ for $n \geq 2$, $n \in N$, then $f_{2008}(x) + f_{2009}(x) =$

(1) $\frac{2x^2+5}{2x-3}$ (2) $\frac{x^2+5}{2x-3}$

(3) $\frac{2x^2-5}{2x-3}$ (4) $\frac{x^2-5}{2x-3}$

8. Consider all functions $f : \{1, 2, 3, 4\} \to \{1, 2, 3, 4\}$ which are one-one, onto and satisfy the following property: if $f(k)$ is odd then $f(k + 1)$ is even, $k = 1, 2, 3$.
The number of such functions is

(1) 4 (2) 8

(3) 12 (4) 16

9. If $h(x) = \left[\ln\frac{x}{e}\right] + \left[\ln\frac{e}{x}\right]$, where [.] denotes greatest integer function, then which of the following are true?

(1) Range of $h(x)$ is $\{-1, 0\}$

(2) If $h(x) = 0$, then x must be irrational

(3) If $h(x) = -1$, then x can be rational as well as irrational

(4) $h(x)$ is periodic function

10. Let $f(x)$ be a real valued continuous function such that

$f(0) = \frac{1}{2}$ and $f(x + y) = f(x)f(a - y) + f(y)f(a - x)$ $\forall$ $x, y \in R$,

then for some real a

(1) $f(x)$ is a periodic function (2) $f(x)$ is a constant function

(3) $f(x) = \frac{1}{2}$ (4) $f(x) = \frac{\cos x}{2}$

11. If the domain of $f(x) = \frac{1}{\pi}\cos^{-1}\left[\log_3\left(\frac{x^2}{3}\right)\right]$ where, $x > 0$ is $[a, b]$ and the range of $f(x)$ is $[c, d]$, then

(1) a, b are the roots of the equation $x^4 - 3x^3 - x + 3 = 0$

(2) a, b are the roots of the equation $x^4 - x^3 + x^2 - 2x + 1 = 0$

(3) $a^3 + d^3 = 1$

(4) $a^2 + b^2 + c^2 + d^2 = 11$

Paragraph for Q. Nos. 12 and 13

Let $f(x)$ be a continuous function (define for all x) which satisfies $f^3(x) - 5f^2(x) + 10f(x) - 12 \geq 0$, $f^2(x) - 4f(x) + 3 \geq 0$ and $f^2(x) - 5f(x) + 6 \leq 0$

12. If distinct positive number b_1, b_2 and b_3 are in G.P. then $f(1) + \ln b_1$, $f(2) + \ln b_2$, $f(3) + \ln b_3$ are in

(1) A.P. (2) G.P.

(3) H.P. (4) A.G.P.

13. The equation of tangent that can be drawn from (2, 0) on the curve $y = x^2 f(\sin x)$ is

(1) $y = 24(x + 2)$ (2) $y = 12(x + 2)$

(3) $y = 24(x - 2)$ (4) $y = 12(x - 2)$

14. If $x, y, z \in R$ satisfies the system of equations $x + [y] + \{z\} = 12.7$, $[x] + \{y\} + z = 4.1$ and $\{x\} + y + [z] = 2$ (where {.} and [.] denote the fractional and integral part respectively), then match the following.

Column-I	Column-II
(A) $\{x\} + \{y\} =$	(P) 7.7
(B) $[z] + [x] =$	(Q) 1.1
(C) $x + \{z\} =$	(R) 1
(D) $z + [y] - \{x\} =$	(S) 3
	(T) 4

15. Let $f(x)$ be a polynomial of degree 6 with leading coefficient 2009. Suppose further, that $f(1) = 1$, $f(2) = 3$, $f(3) = 5$, $f(4) = 7$, $f(5) = 9$, $f'(2) = 2$, then the sum of all the digits of $f(6)$ is

16. Let $f(x) = \dfrac{x+5}{\sqrt{x^2+1}} \forall\, x \in R$, then the smallest integral value of k for which $f(x) \le k \forall\, x \in R$ is

17. If range of $f(x) = \dfrac{(\ln x)(\ln x^2) + \ln x^3 + 3}{\ln^2 x + \ln x^2 + 2}$ can be expressed as $\left[\dfrac{a}{b}, \dfrac{c}{d}\right]$ where a, b, c and d are prime numbers (not necessarily distinct) then find the value of $\dfrac{(a+b+c+d)}{2}$.

18. The value of $\lim\limits_{x\to 0} \dfrac{(\sin x - \tan x)^2 - (1-\cos 2x)^4 + x^5}{7(\tan^{-1} x)^7 + (\sin^{-1} x)^6 + 3\sin^5 x}$ equal to

(1) 0 (2) 1

(3) 2 (4) $\dfrac{1}{3}$

19. $f(x) = \lim\limits_{n\to\infty} \dfrac{x^n + \left(\dfrac{\pi}{3}\right)^n}{x^{n-1} + \left(\dfrac{\pi}{3}\right)^{n-1}}$, ($n$ is an even integer), then which of the following is incorrect?

(1) If $f : \left[\dfrac{\pi}{3}, \infty\right) \to \left[\dfrac{\pi}{3}, \infty\right)$, then function is invertible

(2) $f(x) = f(-x)$ has infinite number of solutions

(3) $f(x) = |f(x)|$ has infinite number of solutions

(4) $f(x)$ is one-one function for all $x \in R$

20. The value of $\lim\limits_{x\to 0}\left[(1-2x)^n \sum\limits_{r=0}^{n} {}^nC_r \left(\dfrac{x+x^2}{1-2x}\right)^r\right]^{1/x}$ is

(1) e^n

(2) e^{-n}

(3) e^{3n}

(4) e^{-3n}

21. If $\lim_{x\to c^-}\{\ln x\}$ and $\lim_{x\to c^+}\{\ln x\}$ exists finitely but they are not equal (where {.} denotes fractional part function), then

(1) 'c' can take only rational values

(2) 'c' can take only irrational values

(3) 'c' can take infinite values in which only one is irrational

(4) 'c' can take infinite values in which only one is rational

22. $\lim_{n\to\infty}\sum_{k=1}^{n}\left(\sin\frac{\pi}{2k}-\cos\frac{\pi}{2k}-\sin\left(\frac{\pi}{2(k+2)}\right)+\cos\frac{\pi}{2(k+2)}\right)=$

(1) 0 (2) 1

(3) 2 (4) 3

23. Consider the sequence

$$u_n=\sum_{r=1}^{n}\frac{r}{2^r},\ n\geq 1$$

Then the limit of u_n as $n\to\infty$ is

(1) 1 (2) e

(3) $\frac{1}{2}$ (4) 2

24. The value of $\lim_{n\to\infty}\left(\frac{n!}{n^n}\right)^{\frac{3n^3+4}{4n^4-1}}$, $n\in N$ is equal to

(1) $\left(\frac{1}{e}\right)^{3/4}$ (2) $e^{3/4}$

(3) e^{-1} (4) 0

25. Let $\lim_{x\to\infty}(2^x+a^x+e^x)^{1/x}=L$

which of the following statement(s) is(are) correct?

(1) If $L=a(a>0)$, then the range of a is $[e,\infty)$ (2) If $L=2e(a>0)$, then the range of a is $\{2e\}$

(3) If $L=e(a>0)$, then the range of a is $(0,e]$ (4) If $L=2a$ $(a>1)$, then the range of a is $\left(\frac{e}{2},\infty\right)$

26. If $f(x)=\lim_{n\to\infty}x\left(\frac{3}{2}+[\cos x]\left(\sqrt{n^2+1}-\sqrt{n^2-3n+1}\right)\right)$ where $[y]$ denotes largest integer $\leq y$, then identify the correct statement(s).

(1) $\lim_{x\to 0}f(x)=0$ (2) $\lim_{x\to\frac{\pi}{2}}f(x)=\frac{3\pi}{4}$

(3) $f(x)=\frac{3x}{2}\ \forall\ x\in\left[0,\ \frac{\pi}{2}\right]$ (4) $f(x)=0\ \forall\ x\in\left(\frac{\pi}{2},\ \frac{3\pi}{2}\right)$

27. If $\alpha, \beta \in \left(-\frac{\pi}{2}, 0\right)$ such that $(\sin\alpha + \sin\beta) + \frac{\sin\alpha}{\sin\beta} = 0$ and $(\sin\alpha + \sin\beta)\frac{\sin\alpha}{\sin\beta} = -1$ and $\lambda = \lim_{n\to\infty} \frac{1+(2\sin\alpha)^{2n}}{(2\sin\beta)^{2n}}$ then

(1) $\alpha = -\frac{\pi}{6}$ (2) $\lambda = 2$

(3) $\alpha = -\frac{\pi}{3}$ (4) $\lambda = 1$

28. Find $\lim_{x\to\alpha^+}\left[\frac{\min(\sin x, \{x\})}{x-1}\right]$ where α is root of equation $\sin x + 1 = x$ (here [.] represent greatest integer and {.} represent fractional part function)

29. Number of points of discontinuity of $f(x) = \left\{\frac{x}{5}\right\} + \left[\frac{x}{2}\right]$ in $x \in [0, 100]$ is/are (where [.] denotes greatest integer function and {.} denotes fractional part function)

(1) 50

(2) 51

(3) 52

(4) 61

30. f is a differentiable function such that $x = f(t^2)$, $y = f(t^3)$ and $f'(1) \neq 0$ if $\left(\frac{d^2y}{dx^2}\right)_{t=1} =$

(1) $\frac{3}{4}\left(\frac{f''(1)+f'(1)}{(f'(1))^2}\right)$ (2) $\frac{3}{4}\left(\frac{f'(1)\cdot f''(1)-f''(1)}{(f'(1))^2}\right)$

(3) $\frac{4}{3}\frac{f''(1)}{(f'(1))^2}$ (4) $\frac{4}{3}\left(\frac{f'(1)f''(1)-f''(1)}{(f'(1))^2}\right)$

31. Let $g(x)$ be the inverse of $f(x)$ such that $f'(x) = \frac{1}{1+x^5}$, then $\frac{d^2(g(x))}{dx^2}$ is equal to

(1) $\frac{1}{1+(g(x))^5}$ (2) $\frac{g'(x)}{1+(g(x))^5}$

(3) $5(g(x))^4(1+(g(x))^5)$ (4) $1+(g(x))^5$

32. If $f(x) = \lim_{n\to\infty}\left(\prod_{i=1}^{n}\cos\left(\frac{x}{2^i}\right)\right)$ then $f'(x)$ is equal to

(1) $\frac{\sin x}{x}$ (2) $\frac{x}{\sin x}$

(3) $\frac{x\cos x - \sin x}{x^2}$ (4) $\frac{\sin x - x\cos x}{\sin^2 x}$

33. Let $f_1(x) = e^x$ and $f_{n+1}(x) = e^{f_n(x)}$ for any $n \geq 1$, $n \in N$. Then for any fixed n, the value of $\frac{d}{dx} f_n(x)$ equals

(1) $f_n(x)$

(2) $f_n(x) f_{n-1}(x) \ldots\ldots f_2(x) f_1(x)$

(3) $f_n(x) f_{n-1}(x)$

(4) $f_n(x) f_{n-1}(x) \ldots\ldots f_2(x) f_1(x) e^x$

34. Number of points where $f(x) = \begin{cases} \max(|x^2 - x - 2|, x^2 - 3x) & ; \quad x \geq 0 \\ \max(\ln(-x), e^x) & ; \quad x < 0 \end{cases}$ is non-differentiable will be

(1) 1 (2) 2

(3) 3 (4) More than 3

35. Let $f : R \to R$ is not identically zero, differentiable function and satisfy the equations $f(xy) = f(x)\, f(y)$ and $f(x + z) = f(x) + f(z)$, then $f(5)$ =

(1) 3 (2) 5

(3) 10 (4) 15

36. If $f(x) = [x] \tan(\pi x)$ then $f'(k^+)$ is equal to ($k \in I$ and [.] dentoes greatest integer frunction)

(1) $(k-1)\pi(-1)^k$ (2) $k\pi$

(3) $k\pi(-1)^{k+1}$ (4) $(k-1)\pi(-1)^{k+1}$

37. If $f(x) = (4 + x)^n$, $n \in N$ and $f^r(0)$ represents the rth derivative of $f(x)$ at $x = 0$, then the value of $\sum_{r=0}^{\infty} \frac{f^r(0)}{r!}$ is equal to

(1) 2^n (2) 3^n

(3) 5^n (4) 4^n

38. If $f(x) = \sqrt{\frac{1 + \sin^{-1} x}{1 - \tan^{-1} x}}$; then $f'(0)$ is equal to

(1) 4 (2) 3

(3) 2 (4) 1

39. If $f(x) = \tan^{-1}(\text{sgn}(x^2 - \lambda x + 1))$ has exactly one point of discontinuity, then the value of λ can be

(1) 1 (2) –1

(3) 2 (4) –2

40. Let f be a differentiable function satisfying $f'(x) = f'(-x)\ \forall\ x \in R$. Then

(1) If $f(1) = f(2)$, then $f(-1) = f(-2)$

(2) $\frac{1}{2} f(x) + \frac{1}{2} f(y) = f\left(\frac{1}{2}(x + y)\right)$ for all real values of x, y

(3) Let $f(x)$ be an even function, then $f(x) = 0\ \forall\ x \in R$

(4) $f(x) + f(-x) = 2f(0)\ \forall\ x \in R$

41. Let $f(x) = (x^2 - 3x + 2)(x^2 + 3x + 2)$ and α, β, γ satisfy $\alpha < \beta < \gamma$ are the roots of $f'(x) = 0$ then which of the following is/are correct ([.] denotes greatest integer function)?

(1) $[\alpha] = -2$

(2) $[\beta] = -1$

(3) $[\beta] = 0$

(4) $[\alpha] = 1$

42. Let $f(x) = x + (1-x)x^2 + (1-x)(1-x^2)x^3 + ... + (1-x)(1-x^2).....(1-x^{n-1})x^n;\ (n \geq 4)$ then

(1) $f(x) = -\prod_{r=1}^{n}(1-x^r)$

(2) $f(x) = 1-\prod_{r=1}^{n}(1-x^r)$

(3) $f'(x) = (1-f(x))\left(\sum_{r=1}^{n}\frac{r\,x^{r-1}}{(1-x^r)}\right)$

(4) $f'(x) = f(x)\left(\sum_{r=1}^{n}\frac{r\,x^{r-1}}{(1-x^r)}\right)$

43. $f(x)$ is differentiable function satisfying the relationship $f^2(x) + f^2(y) + 2(xy - 1) = f^2(x + y)\ \forall\ x, y \in R$

Also $f(x) > 0\ \forall\ x \in R$ and $f(\sqrt{2}) = 2$. Then which of the following statement(s) is/are correct about $f(x)$?

(1) $[f(3)] = 3$ ([.] denotes greatest integer function)

(2) $f(\sqrt{7}) = 3$

(3) $f(x)$ is even

(4) $f'(0) = 0$

Paragraph for Q. Nos. 44 and 45

Suppose a function $f(x)$ satisfies the following conditions

$$f(x+y) = \frac{f(x)+f(y)}{1+f(x)f(y)},\ \forall\ x, y \in R \text{ and } f'(0) = 1$$

Also $-1 < f(x) < 1,\ \forall\ x \in R$

44. $f(x)$ increases in the complete interval

(1) $(-\infty, -1) \cup (-1, 0) \cup (0, 1) \cup (1, \infty)$

(2) $(-\infty, \infty)$

(3) $(-\infty, -1) \cup (-1, 0)$

(4) $(0, 1) \cup (1, \infty)$

45. The value of the limit $\lim_{x\to\infty}(f(x))^x$ is

(1) 0 (2) 1

(3) e (4) e^2

46. Let $f(x)=\begin{cases}[x] & , \quad -2\le x<0\\ |x| & , \quad 0\le x\le 2\end{cases}$

(where [.] denotes the greatest integer function) $g(x) = \sec x$, $x\in R-(2n+1)\frac{\pi}{2}$, $n\in I$

Match the following statements in Column-I with their values in Column-II in the interval $\left(-\frac{3\pi}{2}, \frac{3\pi}{2}\right)$.

Column-I	Column-II
(A) Abscissa of points where limit of $fog(x)$ exist is/are	(P) –1
(B) Abscissa of points in domain of $gof(x)$, where limit of $gof(x)$ does not exist is/are	(Q) π
(C) Abscissa of points of discontinuity of $fog(x)$ is/are	(R) $\frac{5\pi}{6}$
(D) Abscissa of points of differentiability of $fog(x)$ is/are	(S) $-\pi$
	(T) 0

47. Let $f(x) = x^2 + ax + 3$ and $g(x) = x + b$, where $F(x)=\lim_{n\to\infty}\frac{f(x)+(x^2)^n g(x)}{1+(x^2)^n}$. If $F(x)$ is continuous at $x = 1$ and $x = -1$ then find the value of $(a^2 + b^2)$.

48. The number of non-differentiability points of function $f(x)=\min\left([x],\ \{x\},\ \left|x-\frac{3}{2}\right|\right)$ for $x\in(0, 2)$, where [.] and {.} denote greatest integer function and fractional part function respectively.

49. Let A be the point where the curve $5\alpha^2x^3 + 10\alpha x^2 + x + 2y - 4 = 0$ ($\alpha \in R$, $\alpha \neq 0$) meets the y-axis, then the equation of tangent to the curve at the point where normal at A meets the curve again, is

(1) $x - \alpha y + 2\alpha = 0$

(2) $\alpha x + y - 2 = 0$

(3) $2x - y + 2 = 0$

(4) $x + 2y - 4 = 0$

50. The number of critical points of $f(x)=\left(\int_0^x(\cos^2 t-\sqrt[3]{t})dt\right)+\frac{3}{4}x^{4/3}-\frac{x+1}{2}$ in $[0, 6\pi]$ is

(1) 10

(2) 8

(3) 6

(4) 12

51. Let for the function $f(x)=\begin{bmatrix}\cos^{-1}x & ; & -1\le x\le 0\\ mx+c & ; & 0<x\le 1\end{bmatrix}$

Lagrange's mean value theorem is applicable in [–1, 1] then ordered pair (m, c) is

(1) $\left(1,\ -\frac{\pi}{2}\right)$ (2) $\left(1,\ \frac{\pi}{2}\right)$

(3) $\left(-1,\ -\frac{\pi}{2}\right)$ (4) $\left(-1,\ \frac{\pi}{2}\right)$

52. If m and n are positive integers and $f(x)=\int_1^x (t-a)^{2n}(t-b)^{2m+1}dt,\ a\neq b,$ then

(1) $x = b$ is a point of local minimum

(2) $x = b$ is a point of local maximum

(3) $x = a$ is a point of local minimum

(4) $x = a$ is a point of local maximum

53. For any real number b, let $f(b)$ denotes the maximum of $\left|\sin x+\frac{2}{3+\sin x}+b\right| \forall\ x\in R$. Then the minimum value of $f(b)\forall\, b\in R$ is

(1) $\frac{1}{2}$ (2) $\frac{3}{4}$

(3) $\frac{1}{4}$ (4) 1

54. Let $f : [0, 8] \to R$ be differentiable function such that $f(0) = 0$, $f(4) = 1$, $f(8) = 1$, then which of the following hold(s) good?

(1) There exist some $c_1\in(0,\ 8)$ where $f'(c_1)=\frac{1}{4}$

(2) There exist some $c\in(0,\ 8)$ where $f'(c)=\frac{1}{12}$

(3) There exist $c_1,\ c_2\in[0,\ 8]$ where $8f'(c_1)f(c_2)=1$

(4) There exist some $\alpha,\ \beta\in(0,\ 2)$ such that $\int_0^8 f(t)dt=3(\alpha^2 f(\alpha^3)+\beta^2 f(\beta^3))$

55. Let $f(x)$ be twice differentiable function such that $f''(x)>0$ in [0, 2]. Then

(1) $f(0) + f(2) = 2f(c)$, for at least one c, $c \in (0, 2)$

(2) $f(0) + f(2) < 2f(1)$

(3) $f(0) + f(2) > 2f(1)$

(4) $2f(0)+f(2)>3f\left(\frac{2}{3}\right)$

56. Let $F(x) = (f(x))^2 + (f'(x))^2$, $F(0) = 6$, where $f(x)$ is a thrice differentiable function such that $|f(x)| \le 1 \ \forall \ x \in [-1, 1]$, then choose the correct statement(s)

(1) There is at least one point in each of the intervals $(-1, 0)$ and $(0, 1)$ where $|f'(x)| \le 2$

(2) There is at least one point in each of the intervals $(-1, 0)$ and $(0, 1)$ where $F(x) \le 5$

(3) There is no point of local maxima of $F(x)$ in $(-1, 1)$

(4) For some $c \in (-1, 1)$, $F(c) \ge 6$, $F'(c) = 0$ and $F''(c) \le 0$

57. If $(f(x)-1)(x^2+x+1)^2 - (f(x)+1)(x^4+x^2+1) = 0$

$\forall \ x \in R - \{0\}$ and $f(x) \ne \pm 1$, then which of the following statement(s) is/are correct?

(1) $|f(x)| \ge 2 \forall x \in R - \{0\}$

(2) $f(x)$ has a local maximum at $x = -1$

(3) $f(x)$ has a local minimum at $x = 1$

(4) $\int_{-\pi}^{\pi} (\cos x) f(x) dx = 0$

Paragraph for Q. Nos. 58 to 60

The differentiable function $y = f(x)$ has a property that the chord joining any two points $A(x_1, f(x_1))$ and $B(x_2, f(x_2))$ always intersects y-axis at $(0, 2\ x_1\ x_2)$. Given that $f(1) = -1$, then

58. $\int_0^{1/2} f(x) dx$ is equal to

(1) $\frac{1}{6}$ (2) $\frac{1}{8}$

(3) $\frac{1}{12}$ (4) $\frac{1}{24}$

59. The largest interval in which $f(x)$ is monotonically increasing, is

(1) $\left(-\infty, \frac{1}{2}\right]$ (2) $\left[\frac{-1}{2}, \infty\right)$

(3) $\left(-\infty, \frac{1}{4}\right]$ (4) $\left[\frac{-1}{4}, \infty\right)$

60. In which of the following intervals, the Rolle's theorem is applicable to the functionn $F(x) = f(x) + x$?

(1) $[-1, 0]$ (2) $[0, 1]$

(3) $[-1, 1]$ (4) $[0, 2]$

61. $f(x) = \max |2 \sin y - x|$ where $y \in R$ then determine the minimum value of $f(x)$.

62. It is given that $f(x)$ is defined on R satisfying $f(1) = 1$ and for $\forall \ x \in R$,

$f(x + 5) \ge f(x) + 5$ and $f(x + 1) \le f(x) + 1$. If $g(x) = f(x) + 1 - x$, then $g(2002) =$

63. If $I_1 = \int_0^1 \frac{1+x^8}{1+x^4}dx$ and $I_2 = \int_0^1 \frac{1+x^9}{1+x^3}dx$, then

(1) $I_1 > 1,\ I_2 < 1$ (2) $I_1 < 1,\ I_2 > 1$

(3) $1 < I_1 < I_2$ (4) $I_2 < I_1 < 1$

64. If $f(\theta) = \frac{4}{3}(1-\cos^6\theta - \sin^6\theta)$, then $\lim_{n\to\infty} \frac{1}{n}\left[\sqrt{f\left(\frac{1}{n}\right)} + \sqrt{f\left(\frac{2}{n}\right)} + \sqrt{f\left(\frac{3}{n}\right)} + ... + \sqrt{f\left(\frac{n}{n}\right)}\right] =$

(1) $\frac{1-\cos 1}{2}$ (2) $1 - \cos 2$

(3) $\frac{\sin 2}{2}$ (4) $\frac{1-\cos 2}{2}$

65. Evaluate $\int \frac{\left(\sqrt[3]{x+\sqrt{2-x^2}}\right)\left(\sqrt[6]{1-x\sqrt{2-x^2}}\right)dx}{\sqrt[3]{1-x^2}}$; $x \in (0,\ 1)$

(1) $2^{\frac{1}{6}}x + C$ (2) $2^{\frac{1}{12}}x + C$

(3) $2^{\frac{1}{3}}x + C$ (4) $2^{\frac{1}{8}}x + C$

66. Let f is a function, continuous on [0, 1] such that $f(x) \le \sqrt{5}\ \forall x \in [0,\ 1]$ and $f(x) \le \frac{2}{x} \forall\ x \in \left[\frac{1}{2},\ 1\right]$ then the smallest a for which $\int_0^1 f(x)dx \le a$ holds for all f is

(1) $\sqrt{5}$ (2) $\frac{\sqrt{5}}{2} + 2\ln 2$

(3) $2 + \ln\left(\frac{\sqrt{5}}{2}\right)$ (4) $2 + 2\ln\left(\frac{\sqrt{5}}{2}\right)$

67. Value of $\lim_{h\to 0} \frac{\int_0^{\pi+he^{-1/h}} x^2e^{-x^2}dx - \int_0^{\pi} x^2e^{-x^2}dx}{he^{-1/h}}$ is equal to

(1) $\pi(1-\pi^2)e^{-\pi^2}$

(2) $2\pi(1-\pi^2)e^{-\pi^2}$

(3) $\pi(1-\pi)e^{-\pi}$

(4) $\pi^2e^{-\pi^2}$

68. If $\int_0^1 \left(\sum_{r=1}^{2013} \frac{x}{x^2+r^2}\right)\left(\prod_{r=1}^{2013}(x^2+r^2)\right)dx = \frac{1}{2}\left[\left(\prod_{r=1}^{2013}(1+r^2)\right)-k^2\right]$ then k =

(1) 2013 (2) 2013!

(3) 2013^2 (4) 2013^{2013}

69. The value of $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{n-k}{n^2}\cos\frac{4k}{n}$ equals

(1) $\frac{1}{4}\sin 4+\frac{1}{16}\cos 4-\frac{1}{16}$ (2) $\frac{1}{4}\sin 4-\frac{1}{16}\cos 4+\frac{1}{16}$

(3) $\frac{1}{16}(1-\sin 4)$ (4) $\frac{1}{16}(1-\cos 4)$

70. If $A=\int_0^1 \prod_{r=1}^{2014}(r-x)dx$ and $B=\int_0^1 \prod_{r=0}^{2013}(r+x)dx$, then

(1) $A = 2B$ (2) $2A = B$

(3) $A + B = 0$ (4) $A = B$

71. The value of function $f(x)=1+x+\int_1^x(\ln^2 t+2\ln t)dt$ where $f'(x)$ vanishes is

(1) $\frac{1}{e}$ (2) 0

(3) $\frac{2}{e}$ (4) $1+\frac{2}{e}$

72. $\lim_{n\to\infty}\frac{1}{n^3}\left(\sqrt{n^2+1}+2\sqrt{n^2+2^2}+.....+n\sqrt{(n^2+n^2)}\right)=$

(1) $\frac{3\sqrt{2}-1}{2}$ (2) $\frac{2\sqrt{2}-1}{3}$

(3) $\frac{3\sqrt{3}-1}{3}$ (4) $\frac{4\sqrt{2}-1}{2}$

73. The value of $\int_0^1 \lim_{n\to\infty}\sum_{k=0}^{n}\frac{x^{k+2}2^k}{k!}dx$ is

(1) $e^2 - 1$ (2) 2

(3) $\frac{e^2-1}{2}$ (4) $\frac{e^2-1}{4}$

74. $\int \frac{dx}{(1+\sqrt{x})^8} = -\frac{1}{3(1+\sqrt{x})^{k_1}} + \frac{2}{7(1+\sqrt{x})^{k_2}} + C$, then

(1) $k_1 = 5$ (2) $k_1 = 6$

(3) $k_2 = 7$ (4) $k_2 = 8$

75. Let $J = \int_{-1}^{2} \left(\cot^{-1}\frac{1}{x} + \cot^{-1} x \right) dx$, $K = \int_{-2\pi}^{7\pi} \frac{\sin x}{|\sin x|} dx$. Then which of the following alternative(s) is/are correct?

(1) $2J + 3K = 8\pi$ (2) $4J^2 + K^2 = 26\pi^2$

(3) $2J - K = 3\pi$ (4) $\frac{J}{K} = \frac{2}{5}$

Paragraph for Q. Nos. 76 and 77

Let $g(x) = x^C e^{Cx}$ and $f(x) = \int_0^x te^{2t}(1+3t^2)^{1/2} dt$. If $L = \lim_{x\to\infty} \frac{f'(x)}{g'(x)}$ is non-zero finite number then

76. The value of C is

(1) 7 (2) $\frac{3}{2}$

(3) 2 (4) 3

77. The value of L is

(1) $\frac{2}{7}$ (2) $\frac{1}{2}$

(3) $\frac{\sqrt{3}}{4}$ (4) $\frac{\sqrt{3}}{2}$

78. Match Column-I with Column-II.

Column-I	Column-II
(A) $\lim_{n\to\infty} 4\left[\frac{e^{\frac{1}{n}}}{n^2} + \frac{2}{n^2}e^{\frac{2}{n}} + \frac{3}{n^2}e^{\frac{3}{n}} + \ldots \frac{1}{n}e\right] =$	(P) 0
(B) $\int_0^1 \ln\left(\frac{1}{x} - 1\right) dx =$	(Q) 1
(C) $\int_0^{10\pi} \left(\lim_{x\to y} \left(\frac{\sin x - \sin y}{x - y} \right) \right) dy =$	(R) 2
(D) $\int_0^{\infty} \frac{\ln\left(x + \frac{1}{x}\right) dx}{(1+x^2)} = \frac{\pi}{2}\ln a$, then $a =$	(S) 4
	(T) 5

79. Let $\int_1^{\sqrt{3}} \left(x^{2x^2+1} + \ln(x^{2x^{2x^2+1}}) \right) dx = N$. Find the value of $(N - 6)$.

80. If maximum value of $\int_0^1 (f(x))^3 dx$ under the condition $-1 \le f(x) \le 1$; $\int_0^1 f(x)dx = 0$ is $\frac{p}{q}$

(where p and q are relatively prime positive integers). Find $p + q$.

81. Find the value of $\lim_{n\to\infty} \frac{1}{\sqrt{n}}\left(1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + ... + \frac{1}{\sqrt{n}}\right)$.

82. Let $\lim_{n\to\infty} n^{-\frac{1}{2}\left(1+\frac{1}{n}\right)} \cdot (1^1 \cdot 2^2 \cdot 3^3 \cdot .....n^n)^{\frac{1}{n^2}} = e^{\frac{-p}{q}}$

where p and q are relative prime positive integers. Find the value of $|p + q|$.

83. The area enclosed by the curve $[x + 3y] = [x - 2]$ where $x \in [3, 4)$ is
(where [.] denotes greatest integer function)

(1) $\frac{2}{3}$ (2) $\frac{1}{3}$

(3) $\frac{1}{4}$ (4) 1

84. For $0 < r < 1$, let n_r denotes the line that is normal to the curve $y = x^r$ at the point (1, 1). Let S_r denotes the region in the first quadrant bounded by the curve $y = x^r$; the x-axis and the line n_r. Then the value of r that minimizes the area of S_r is

(1) $\frac{1}{\sqrt{2}}$ (2) $\sqrt{2} - 1$

(3) $\frac{\sqrt{2}-1}{2}$ (4) $\sqrt{2} - \frac{1}{2}$

85. If $f(x) = \min\left[x^2,\ \sin\frac{x}{2},\ (x - 2\pi)^2 \right]$, the area bounded by the curve $y = f(x)$, x-axis, $x = 0$ and $x = 2\pi$ is given by

(**Note :** x_1 is the point of intersection of the curves x^2 and $\sin\frac{x}{2}$; x_2 is the point of intersection of the curves $\sin\frac{x}{2}$ and $(x - 2\pi)^2$)

(1) $\int_0^{x_1} \left(\sin\frac{x}{2}\right) dx + \int_{x_1}^{\pi} x^2 dx + \int_{\pi}^{x_2} (x - 2\pi)^2 dx + \int_{x_2}^{2\pi} \left(\sin\frac{x}{2}\right) dx$

(2) $\int_0^{x_1} x^2 dx + \int_{x_1}^{x_2} \left(\sin\frac{x}{2}\right) dx + \int_{x_2}^{2\pi} (x - 2\pi)^2 dx$, where $x_1 \in \left(0,\ \frac{\pi}{3}\right)$ and $x_2 \in \left(\frac{5\pi}{3},\ 2\pi\right)$

(3) $\int_0^{x_1} x^2 dx + \int_{x_1}^{x_2} \sin\left(\frac{x}{2}\right) dx + \int_{x_2}^{2\pi} (x - 2\pi)^2 dx$, where $x_1 \in \left(\frac{\pi}{3},\ \frac{\pi}{2}\right)$ and $x_2 \in \left(\frac{3\pi}{2},\ 2\pi\right)$

(4) $\int_0^{x_1} x^2 dx + \int_{x_1}^{x_2} \sin\left(\frac{x}{2}\right) dx + \int_{x_2}^{2\pi} (x - 2\pi)^2 dx$, where $x_1 \in \left(\frac{\pi}{2},\ \frac{2\pi}{3}\right)$ and $x_2 \in (\pi,\ 2\pi)$

Paragraph for Q. Nos. 86 to 88

Let $f : A \to B\ f(x) = \dfrac{x+a}{bx^2+cx+2}$, where A represent domain set and B represent range set of function $f(x)$, $a, b, c \in R$, $f(-1) = 0$ and $y = 1$ is an asymptote of $y = f(x)$ and $y = g(x)$ is the inverse of $f(x)$.

86. $g(0)$ is equal to

(1) –1 (2) –3

(3) $-\dfrac{5}{2}$ (4) $-\dfrac{3}{2}$

87. Area bounded between the curves $y = f(x)$ and $y = g(x)$ is

(1) $2\sqrt{5}+\ln\left(\dfrac{3-\sqrt{5}}{3+\sqrt{5}}\right)$ (2) $3\sqrt{5}+2\ln\left(\dfrac{3+\sqrt{5}}{3-\sqrt{5}}\right)$

(3) $3\sqrt{5}+4\ln\left(\dfrac{3-\sqrt{5}}{3+\sqrt{5}}\right)$ (4) $3\sqrt{5}+2\ln\left(\dfrac{3-\sqrt{5}}{3+\sqrt{5}}\right)$

88. Area of region enclosed by asymptotes of curves $y = f(x)$ and $y = g(x)$ is

(1) 4 (2) 9

(3) 12 (4) 25

89. Match Column-I with Column-II.

Column-I	**Column-II**
(A) Area of region formed by points (x, y) satisfying $[x]^2 = [y]^2$ for $0 \le x \le 4$ is equal to (where [ ] denotes greatest integer function)	(P) 48
(B) The area of region formed by points (x, y) satisfying $x + y \le 6$, $x^2 + y^2 \le 6y$ and $y^2 \le 8x$ is $\dfrac{k\pi-2}{12}$, then k =	(Q) 27
(C) The area in the first quadrant bounded by the curve $y = \sin x$ and the line $\dfrac{2y-1}{\sqrt{2}-1} = \dfrac{2}{\pi}(6x-\pi)$ is $\left[\dfrac{\sqrt{3}-\sqrt{2}}{2}-\dfrac{(\sqrt{2}+1)\pi}{k}\right]$, then k =	(E) 7
(D) If the area bounded by the graph of $y = xe^{-ax}$ $(a > 0)$ and the abscissa axis is $\dfrac{1}{9}$ then the value of a is equal to	(S) 4
	(T) 3

90. If the area bounded by circle $x^2 + y^2 = 4$, the parabola $y = x^2 + x + 1$ and the curve $y = \left[\sin^2\frac{x}{4} + \cos\frac{x}{4}\right]$, (where [ ] denotes the greatest integer function) and x-axis is $\left(\sqrt{3} + \frac{2\pi}{3} - \frac{1}{k}\right)$, then the numerical quantity k should be

91. The differential equation satisfied by family of curves $y = Ae^x + Be^{3x} + Ce^{5x}$ where A, B, C are arbitrary constants is

(1) $\frac{d^3y}{dx^3} - 9\frac{d^2y}{dx^2} + 23\frac{dy}{dx} + 15y = 0$ (2) $\frac{d^3y}{dx^3} + 9\frac{d^2y}{dx^2} - 23\frac{dy}{dx} - 15y = 0$

(3) $\frac{d^3y}{dx^3} + 9\frac{d^2y}{dx^2} - 23\frac{dy}{dx} + 15y = 0$ (4) $\frac{d^3y}{dx^3} - 9\frac{d^2y}{dx^2} + 23\frac{dy}{dx} - 15y = 0$

92. A function $y = f(x)$ satisfies the differential equation $(x+1)f'(x) - 2(x^2 + x)f(x) = \frac{e^{x^2}}{(x+1)}$; $\forall\ x > -1$. If $f(0) = 5$, then $f(x)$ is

(1) $\left(\frac{3x+5}{x+1}\right)\cdot e^{x^2}$ (2) $\left(\frac{6x+5}{x+1}\right)\cdot e^{x^2}$

(3) $\left(\frac{6x+5}{(x+1)^2}\right)\cdot e^{x^2}$ (4) $\left(\frac{5-6x}{x+1}\right)\cdot e^{x^2}$

93. Let $f(x)$ be differentiable function on the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim_{t\to x}\left(\frac{t^3f(x) - x^3f(t)}{t^2 - x^2}\right) = \frac{1}{2}\forall\ x > 0$, then $f(x)$ is

(1) $\frac{1}{4x} + \frac{3x^2}{4}$ (2) $\frac{3}{4x} + \frac{x^3}{4}$

(3) $\frac{1}{4x} + \frac{3x^3}{4}$ (4) $\frac{1}{4x^3} + \frac{3x}{4}$

94. Solution of the differential equation $(2 + 2x^2\sqrt{y})\,ydx + (x^2\sqrt{y} + 2)x\,dy = 0$ is/are

(1) $xy(x^2\sqrt{y} + 5) = c$ (2) $xy(x^2\sqrt{y} + 3) = c$

(3) $xy(y^2\sqrt{x} + 3) = c$ (4) $xy(y^2\sqrt{x} + 5) = c$

Paragraph for Q. Nos. 95 to 97

Suppose f and g are differentiable functions such that $xg(f(x))f'(g(x))g'(x) = f(g(x))g'(f(x))f'(x)\ \forall\ x \in R$ and f is positive, g is positive $\forall\ x \in R$. Also $\int_0^x f(g(t))dt = \frac{1}{2}(1 - e^{-2x})\forall\ x \in R$, $g(f(0)) = 1$ and $h(x) = \frac{g(f(x))}{f(g(x))}\forall\ x \in R$.

95. The graph of $y = h(x)$ is symmetric with respect to line

(1) $x = -1$ (2) $x = 0$

(3) $x = 1$ (4) $x = 2$

96. The value of $f(g(0)) + g(f(0))$ is equal to

(1) 1 (2) 2

(3) 3 (4) 4

97. The largest possible value of $h(x)\ \forall\ x \in R$ is

(1) 1 (2) $e^{1/3}$

(3) e (4) e^2

98. Let $y = (a\sin x + (b+c)\cos x)e^{x+d}$, where a, b, c and d are parameters represent a family of curves, then differential equation for the given family of curves is given by $y'' - \alpha y' + \beta y = 0$, then $\alpha + \beta =$

99. If $f(x) = \prod_{i=1}^{3}(x - a_i) + \sum_{i=1}^{3} a_i - 3x$ where $a_i < a_{i+1}$ for $i = 1, 2$, then $f(x) = 0$ has

(1) Only one distinct real root

(2) Exactly two distinct real roots

(3) Exactly 3 distinct real roots

(4) 3 Equal real roots

100. The number of integral roots of the equation $x^8 - 24x^7 - 18x^5 + 39x^2 + 1155 = 0$ is

(1) 0 (2) 2

(3) 4 (4) 6

101. If $\tan\theta_i$; $i = 1, 2, 3, 4$ are the roots of equation $x^4 - x^3\sin 2\beta + x^2\cos 2\beta - x\cos\beta - \sin\beta = 0$, then $\tan(\theta_1 + \theta_2 + \theta_3 + \theta_4) =$

(1) $\sin\beta$ (2) $\cos\beta$

(3) $\tan\beta$ (4) $\cot\beta$

102. Let a, b, c be the lengths of the sides of a triangle (no two of them are equal) and $k \in R$. If the roots of the equation $x^2 + 2(a+b+c)x + 6k(ab+bc+ca) = 0$ are real, then

(1) $k < \frac{2}{3}$ (2) $k > \frac{2}{3}$

(3) $k > 1$ (4) $k < \frac{1}{4}$

103. If $\alpha, \beta, \gamma, \delta \in R$ satisfy $\frac{(\alpha+1)^2 + (\beta+1)^2 + (\gamma+1)^2 + (\delta+1)^2}{\alpha+\beta+\gamma+\delta} = 4$

If biquadratic equation $a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 = 0$ has the roots

$\left(\alpha + \frac{1}{\beta} - 1\right)$, $\left(\beta + \frac{1}{\gamma} - 1\right)$, $\left(\gamma + \frac{1}{\delta} - 1\right)$, $\left(\delta + \frac{1}{\alpha} - 1\right)$. Then the value of a_2/a_0 is

(1) 4 (2) –4

(3) 6 (4) –6

104. For real numbers x and y, if $x^2 + xy - y^2 + 2x - y + 1 = 0$, then

(1) y cannot be between 0 and $\frac{8}{5}$

(2) y cannot be between $-\frac{8}{5}$ and $\frac{8}{5}$

(3) y cannot be between $-\frac{8}{5}$ and 0

(4) y cannot be between $-\frac{16}{5}$ and 0

105. Let k be a real number. The minimum number of distinct real roots possible of the equation $(3x^2 + kx + 3)(x^2 + kx - 1) = 0$ is

(1) 0

(2) 2

(3) 3

(4) 4

106. Assume that p is a real number. In order of $\sqrt[3]{x+3p+1} - \sqrt[3]{x} = 1$ to have real solutions, it is necessary that

(1) $p \geq 1/4$

(2) $p \geq -1/4$

(3) $p \geq 1/3$

(4) $p \geq -1/3$

107. Two particles, A and B, are in motion in the xy-plane. Their co-ordinates at each instant of time t ($t \geq 0$) are given by $x_A = t$, $y_A = 2t$, $x_B = 1 - t$ and $y_B = t$. The minimum distance between particles A and B is

(1) $\frac{1}{5}$

(2) $\frac{1}{\sqrt{5}}$

(3) 1

(4) $\sqrt{\frac{2}{3}}$

108. If $a + b + c = 1$, $a^2 + b^2 + c^2 = 9$ and $a^3 + b^3 + c^3 = 1$, then the value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ is

(1) $\frac{2}{3}$

(2) 5

(3) 6

(4) 1

109. If the equation $|\sin x|^2 + |\sin x| + b = 0$ has two distinct roots in $[0, \pi]$; then the number of integers in the range of b is equal to

(1) 0

(2) 1

(3) 2

(4) 3

110. If exactly two integers lie between the roots of equation $x^2 + ax - 1 = 0$. Then integral value(s) of a is/are

(1) –1

(2) –2

(3) 1

(4) 2

111. If all values of x which satisfies the inequality $\log_{(1/3)}(x^2 + 2px + p^2 + 1) \geq 0$ also satisfy the inequality $kx^2 + kx - k^2 \leq 0$ for all real values of k, then all possible values of p lies in the interval

(1) [–1, 1]

(2) [0, 1]

(3) [0, 2]

(4) [–2, 0]

112. Given a, b, c are three distinct real numbers satisfying the inequality $a - 2b + 4c > 0$ and the equation $ax^2 + bx + c = 0$ has no real roots. Then the possible value(s) of $\frac{4a+2b+c}{a+3b+9c}$ is/are

(1) 2 (2) –1

(3) 3 (4) $\sqrt{2}$

113. Let $\alpha, \beta, \gamma, \delta$ are roots of $x^4 - 12x^3 + \lambda x^2 - 54x + 14 = 0$.

If $\alpha + \beta = \gamma + \delta$, then

(1) $\lambda = 45$ (2) $\lambda = -45$

(3) If $\alpha^2 + \beta^2 < \gamma^2 + \delta^2$ then $\frac{\alpha\beta}{\gamma\delta} = \frac{7}{2}$ (4) If $\alpha^2 + \beta^2 < \gamma^2 + \delta^2 \Rightarrow \frac{\alpha\beta}{\gamma\delta} = \frac{2}{7}$

Paragraph for Q. Nos. 114 and 115

The equation $x^4 - 2x^3 - 3x^2 + 4x - 1 = 0$ has four distinct real roots x_1, x_2, x_3, x_4 such that $x_1 < x_2 < x_3 < x_4$ and product of two roots is unity, then

114. $x_1x_2 + x_1x_3 + x_2x_4 + x_3x_4 =$

(1) 0 (2) 1

(3) $\sqrt{5}$ (4) –1

115. $x_2^3 + x_4^3 =$

(1) $\frac{2+5\sqrt{5}}{8}$ (2) –4

(3) $\frac{27\sqrt{5}+5}{4}$ (4) 18

116. Match Column-I with Column-II.

Column-I	**Column-II**
(A) If a, b are the roots of equation $x^2 + ax + b = 0$ ($a, b \in R$), then the number of ordered pair (a, b) is equal to	(P) 1
(B) If $P = \text{cosec}\frac{\pi}{8} + \text{cosec}\frac{2\pi}{8} + \text{cosec}\frac{3\pi}{8} + \text{cosec}\frac{13\pi}{8} + \text{cosec}\frac{14\pi}{8} + \text{cosec}\frac{15\pi}{8}$, and $Q = 8\sin\frac{\pi}{18}\sin\frac{5\pi}{18}\sin\frac{7\pi}{18}$ then $P + Q$ is equal to	(Q) 2
(C) Let a_1, a_2, a_3 ..... be positive terms of a G.P. and a_4, 1, 2, a_{10} are the consecutive terms of another G.P. If $\prod_{i=2}^{12} a_i = 4^{\frac{m}{n}}$ where m and n are coprime, then ($m + n$) equals	(R) 3
(D) For $x, y \in R$, if $x^2 - 2xy + 2y^2 - 6y + 9 = 0$, then the value of $5x - 4y$ is equal to	(S) 15

117. If the equation $(m^2 - 12)x^4 - 8x^2 - 4 = 0$ has no real roots, then the largest value of m is $p\sqrt{q}$ where p, q are coprime natural numbers, then $p + q$ =

118. The number of non-negative integral values of n, $n \leq 10$ so that a root of the equation $n^2 \sin^2 x - 2 \sin x - (2n + 1) = 0$ lies in interval $\left[0, \frac{\pi}{2}\right]$ is

119. The real value of x satisfying $\sqrt[3]{20x + \sqrt[3]{20x + 13}} = 13$ can be expressed as $\frac{a}{b}$ where a and b are relatively prime positive integers. Find the value of b.

120. If $(m + 1)^{th}$, $(n + 1)^{th}$ and $(r + 1)^{th}$ terms of a non-constant A.P. are in G.P. and m, n, r are in H.P., then the ratio of first term of the A.P. to its common difference is

(1) $-\frac{n}{2}$ (2) $-n$

(3) $-2n$ (4) $+n$

121. Let $S_k = 1 + 2 + 3 + ... + k$ and $Q_n = \frac{S_2}{S_2 - 1} \cdot \frac{S_3}{S_3 - 1} \cdot \frac{S_4}{S_4 - 1} \cdots \frac{S_n}{S_n - 1}$, where $k, n \in N$

$\lim_{n \to \infty} Q_n$ =

(1) $\frac{1}{3}$ (2) 1

(3) 3 (4) 0

122. Let α, β be two distinct values of x lying in $[0, \pi]$ for which $\sqrt{5}\sin x$, $10 \sin x$, $10(4 \sin^2 x + 1)$ are 3 consecutive terms of a G.P. Then minimum value of $|\alpha - \beta|$ =

(1) $\frac{\pi}{10}$ (2) $\frac{\pi}{5}$

(3) $\frac{2\pi}{5}$ (4) $\frac{3\pi}{5}$

123. Let $S_k = \sum_{i=0}^{\infty} \frac{1}{(k+1)^i}$, then $\sum_{k=1}^{n} kS_k$ equals

(1) $\frac{n(n+1)}{2}$ (2) $\frac{n(n-1)}{2}$

(3) $\frac{n(n+2)}{2}$ (4) $\frac{n(n+3)}{2}$

124. The number of non-negative integers n satisfying $n^2 = p + q$ and $n^3 = p^2 + q^2$ where p and q are integers

(1) 2 (2) 3

(3) 4 (4) Infinite

125. The absolute term in $P(x)=\sum_{r=1}^{n}\left(x-\frac{1}{r}\right)\left(x-\frac{1}{r+1}\right)\left(x-\frac{1}{r+2}\right)$ as n approaches to infinity is

(1) $\frac{1}{2}$ (2) $\frac{-1}{2}$

(3) $\frac{1}{4}$ (4) $\frac{-1}{4}$

126. $\sum_{r=1}^{10}\frac{r}{1-3r^2+r^4}=$

(1) $-\frac{50}{109}$ (2) $-\frac{54}{109}$

(3) $-\frac{55}{111}$ (4) $-\frac{55}{109}$

127. If a, b, c are 3 distinct numbers in H.P., a, b, $c > 0$, then

(1) $\frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in A.P.

(2) $\frac{b+c}{a}, \frac{c+a}{b}, \frac{a+b}{c}$ are in A.P.

(3) $a^5+c^5 \geq 2b^5$

(4) $\frac{a-b}{b-c}=\frac{a}{c}$

128. Let a_1, a_2, a_3, ... be a sequence of non-zero real numbers which are in A.P. for $k \in N$.

Let $f_k(x) = a_k x^2 + 2a_{k+1}\, x + a_{k+2}$

(1) $f_k(x) = 0$ has real roots for each $k \in N$

(2) Each of $f_k(x) = 0$ has one root in common

(3) Non-common roots of $f_1(x) = 0$, $f_2(x) = 0$, $f_3(x) = 0$, ... form an A.P.

(4) None of these

129. Given $\frac{\underbrace{1000.....01}_{n \text{ zeroes}}}{\underbrace{1000.....01}_{(n+1) \text{ zeroes}}} < \frac{\underbrace{1000.....01}_{\text{m zeroes}}}{\underbrace{1000.....01}_{(m+1) \text{ zeroes}}}$, then which of the following is true

(1) $m + 1 < n$ (2) $m < n$

(3) $m < n + 1$ (4) $m > n + 1$

Paragraph for Q. No.s 130 and 131

Let $f(n)=\sum_{r=2}^{n}\frac{r}{{}^{r}C_2\ {}^{r+1}C_2}$, $a=\lim_{n\to\infty} f(n)$ and $x^2-\left(2a-\frac{1}{2}\right)x+t=0$ has two positive roots α and β.

130. If value of $f(7) + f(8)$ is $\frac{p}{q}$ where p and q are relatively prime, then $(p - q)$ is

(1) 53 (2) 55

(3) 57 (4) 59

131. Minimum value of $\frac{4}{\alpha}+\frac{1}{\beta}$ is

(1) 2 (2) 6

(3) 3 (4) 4

132. Let $f(n)=1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+.....+\frac{1}{n}$ such that $P(n)\, f(n+2) = P(n)\, f(n) + q(n)$. Where $P(n)$, $Q(n)$ are polynomials of least possible degree and $P(n)$ has leading coefficient unity. Then match the following Column-I with Column-II.

Column-I	**Column-II**
(A) $\sum_{n=1}^{m}\frac{p(n)-2}{n}$	(P) $\frac{m(m+1)}{2}$
(B) $\sum_{n=1}^{m}\frac{q(n)-3}{2}$	(Q) $\frac{5m(m+7)}{2}$
(C) $\sum_{n=1}^{m}\frac{p(n)+q^2(n)-11}{n}$	(R) $\frac{3m(m+7)}{2}$
(D) $\sum_{n=1}^{m}\frac{q^2(n)-p(n)-7}{n}$	(S) $\frac{m(m+7)}{2}$

133. Three distinct positive numbers a, b, c are in G.P., while $\log_c a$, $\log_b c$, $\log_a b$ are in A.P. with non-zero common difference d, then $2d$ =

134. Let a and b be positive integers. The value of xyz is 55 and $\frac{343}{55}$ when a, x, y, z, b are in arithmatic and harmonic progression respectively. Find the value of $(a + b)$.

135. If $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$ and vectors $(1, a, a^2)(1, b, b^2)$ and $(1, c, c^2)$ are non-coplanar, then the product abc equals

(1) 2 (2) –1

(3) 1 (4) 0

136. If $D_1 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $D_2 = \begin{vmatrix} a_1+2a_2+3a_3 & 2a_3 & 5a_2 \\ b_1+2b_2+3b_3 & 2b_3 & 5b_2 \\ c_1+2c_2+3c_3 & 2c_3 & 5c_2 \end{vmatrix}$ then $\frac{D_2}{D_1}$ is equal to

(1) 10 (2) –10

(3) 20 (4) –20

137. Let $\det A = \begin{vmatrix} l & m & n \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$ and

if $(l-m)^2 + (p-q)^2 = 9$, $(m-n)^2 + (q-r)^2 = 16$, $(n-l)^2 + (r-p)^2 = 25$, then the value of (det. A)^2 equals

(1) 36 (2) 100

(3) 144 (4) 169

138. Let $D(x) = \begin{vmatrix} x^2+4x-3 & 2x+4 & 13 \\ 2x^2+5x-9 & 4x+5 & 26 \\ 8x^2-6x+1 & 16x-6 & 104 \end{vmatrix} = \alpha x^3 + \beta x^2 + \gamma x + \delta$ then

(1) $\alpha + \beta = 0$

(2) $\beta + \gamma = 0$

(3) $\alpha + \beta + \gamma + \delta = 0$

(4) $\alpha + \beta + \gamma = 0$

139. If the system of equations

$ax + y + 2z = 0$

$x + 2y + z = b$

$2x + y + az = 0$

has no solution then $(a + b)$ can be equal to

(1) –1 (2) 2

(3) 3 (4) 4

140. If $x, y, z \in R$ and $\begin{vmatrix} x & y^2 & z^3 \\ x^4 & y^5 & z^6 \\ x^7 & y^8 & z^9 \end{vmatrix} = 2$ then find the value of

$$\begin{vmatrix} y^5z^6(z^3-y^3) & x^4z^6(x^3-z^3) & x^4y^5(y^3-x^3) \\ y^2z^3(y^6-z^6) & xz^3(z^6-x^6) & xy^2(x^6-y^6) \\ y^2z^3(z^3-y^3) & xz^3(x^3-z^3) & xy^2(y^3-x^3) \end{vmatrix}$$

141. Let $\lim\limits_{n\to\infty} n\sin(2\pi e\lfloor n \rfloor) = k\pi$, where $n \in N$. Find k.

142. The number of points of intersection of the curves represented by

$\arg(z-2-7i) = \cot^{-1}(2)$ and $\arg\left(\dfrac{z-5i}{z+2-i}\right) = \pm\dfrac{\pi}{2}$

(1) 0

(2) 1

(3) 2

(4) More than 2

143. It is given that complex numbers z_1 and z_2 satisfy $|z_1| = 2$ and $|z_2| = 3$. If the included angle of their corresponding vectors is 60°, then $\left|\frac{z_1+z_2}{z_1-z_2}\right|$ can be expressed as $\frac{\sqrt{n}}{7}$, where n is a natural number then n =

(1) 126 (2) 119

(3) 133 (4) 19

144. Let z_1, z_2 and z_3 be three distinct complex numbers, satisfying $|z_1| = |z_2| = |z_3| = 1$. Which of the following is/are true?

(1) If $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$ then $\arg\left(\frac{z-z_1}{z-z_2}\right) > \frac{\pi}{4}$ where $|z| > 1$

(2) $|z_1z_2 + z_2z_3 + z_3z_1| = |z_1 + z_2 + z_3|$

(3) $\text{Im}\left(\frac{(z_1+z_2)(z_2+z_3)(z_3+z_1)}{z_1 \cdot z_2 \cdot z_3}\right) = 0$

(4) If $|z_1 - z_2| = \sqrt{2}|z_1 - z_3| = \sqrt{2}|z_2 - z_3|$, then $\text{Re}\left(\frac{z_3-z_1}{z_3-z_2}\right) = 0$

145. The solutions of the equation $z^4 + 4iz^3 - 6z^2 - 4iz - i = 0$ represent vertices of a convex polygon in the complex plane. The area of the polygon is

(1) $2^{1/2}$ (2) $2^{3/2}$

(3) $2^{5/2}$ (4) $2^{5/4}$

Paragraph for Q. Nos. 146 and 147

In the Argand plane Z_1, Z_2 and Z_3 are respectively the vertices of an isosceles triangle ABC with $AC = BC$ and $\angle CAB = \theta$. If $I(Z_4)$ is the incentre of triangle, then

146. The value of $\left(\frac{AB}{IA}\right)^2\left(\frac{AC}{AB}\right)$ is equal to

(1) $\left|\frac{(Z_2-Z_1)(Z_1-Z_3)}{(Z_4-Z_1)}\right|$ (2) $\left|\frac{(Z_2-Z_1)(Z_3+Z_1)}{(Z_4-Z_1)}\right|$

(3) $\left|\frac{(Z_2-Z_1)(Z_3-Z_1)}{(Z_4-Z_1)^2}\right|$ (4) $\left|\frac{(Z_2+Z_1)(Z_3+Z_1)}{(Z_4+Z_1)}\right|$

147. The value of $(Z_4 - Z_1)^2(1 + \cos\theta)\sec\theta$ is

(1) $(Z_2 - Z_1)(Z_3 - Z_1)$ (2) $\frac{(Z_2-Z_1)(Z_3-Z_1)}{Z_4-Z_1}$

(3) $\frac{(Z_2-Z_1)(Z_3-Z_1)}{(Z_4-Z_1)^2}$ (4) $(Z_2 - Z_1)(Z_3 - Z_1)^2$

148. In a ΔABC, the side lengths BC, CA and AB are consecutive positive integers in increasing order.

Column-I	Column-II
(A) If z_1, z_2 and z_3 be the affixes of vertices A, B and C respectively in argand plane, such that $\left\|\arg\left(\frac{z_1-z_3}{z_2-z_3}\right)\right\| = \left\|2\arg\left(\frac{z_3-z_1}{z_2-z_1}\right)\right\|$, then biggest side of the triangle is	(P) 2
(B) Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be the position vectors of vertices A, B and C respectively. If $(\vec{c}-\vec{a})\cdot(\vec{b}-\vec{c})=0$ then the value of $\|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\|$ equals to	(Q) 3
(C) Let the equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ represent the lines AB and AC respectively and $\left\|\frac{a_1b_2-a_2b_1}{a_1a_2+b_1b_2}\right\| = \frac{4}{3}$ then the value of $s - c$ (where s is the semiperimeter) $a = BC$, $b = CA$, $c = AB$	(R) 4
(D) If the altitudes of ΔABC are in harmonic progression then the side length b can be	(S) 6
	(T) 12

149. Let complex number z satisfy the inequality $2 \le |z| \le 4$. A point P is selected in this region at random. The probability that argument of P lies in the interval $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ is $\frac{1}{k}$, then k =

150. Let $\frac{1}{a_1+\omega}+\frac{1}{a_2+\omega}+\frac{1}{a_3+\omega}+.....+\frac{1}{a_n+\omega}=i$ where $a_1, a_2, a_3, ...a_n \in R$ and ω is imaginary cube root of unity, then evaluate $\sum_{r=1}^{n}\frac{2a_r-1}{a_r^2-a_r+1}$.

151. Let matix $A=\begin{bmatrix} x & 3 & 2 \\ 1 & y & 4 \\ 2 & 2 & z \end{bmatrix}$; if $xyz = 2\lambda$ and $8x + 4y + 3z = \lambda + 28$ then (adj A) A equals

(1) $\begin{bmatrix} \lambda+1 & 0 & 0 \\ 0 & \lambda+1 & 0 \\ 0 & 0 & \lambda+1 \end{bmatrix}$

(2) $\begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$

(3) $\begin{bmatrix} \lambda^{2s} & 0 & 0 \\ 0 & \lambda^2 & 0 \\ 0 & 0 & \lambda^2 \end{bmatrix}$

(4) $\begin{bmatrix} \lambda+2 & 0 & 0 \\ 0 & \lambda+2 & 0 \\ 0 & 0 & \lambda+2 \end{bmatrix}$

152. Let M be a column vector (not null vector) and $A = \frac{MM^T}{M^TM}$ the matrix A is (where M^T is transpose matrix of M)

(1) Idempotant (2) Nilpotent

(3) Involutary (4) Orthogonal

153. If A and B are two orthogonal matrices of order n and det (A) + det (B) = 0, then which of the following must be correct?

(1) $\det(A+B) = \det(A) + \det(B)$ (2) $\det(A+B) = 0$

(3) A and B both are singular matrices (4) $A + B = 0$

154. Match Column-I with Column-II.

	Column-I	**Column-II**
(A)	If P and Q are variable points on $C_1 : x^2 + y^2 = 4$ and $C_2 : x^2 + y^2 - 8x - 6y + 24 = 0$ respectively then the maximum value of PQ, is equal to	(P) 1
(B)	Let P, Q, R be invertible matrices of second order such that $A = PQ^{-1}$, $B = QR^{-1}$, $C = RP^{-1}$, then the value of det. ($ABC + BCA + CAB$) is equal to	(Q) 2
(C)	The perpendicular distance of the point whose position vector is (1, 3, 5) from the line $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$ is equal to	(R) 9
(D)	Let $f(x)$ be a continuous function in [–1, 1] such that $f(x) = \begin{cases} \frac{\ln(px^2+qx+r)}{x^2} & ; \quad -1 \le x < 0 \\ 1 & ; \quad x = 0 \\ \frac{\sin(e^{x^2}-1)}{x^2} & ; \quad 0 < x \le 1 \end{cases}$ then the value of ($p + q + r$), is equal to	(S) 8

155. A and B are two square matrices. Such that $A^2B = BA$ and if $(AB)^{10} = A^k \cdot B^{10}$. Find the value of $k - 1020$.

156. Let $A = \{x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8\}$, $B = \{y_1, y_2, y_3, y_4\}$. The total number of function $f : A \to B$ that are onto and there are exactly three elements x in A such that $f(x) = y_1$ is

(1) 11088 (2) 10920

(3) 13608 (4) None of these

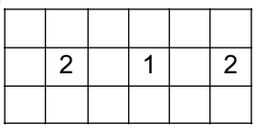
157. The value of $\left[\frac{2009! + 2006!}{2008! + 2007!}\right] =$

([.] denotes greatest integer function)

(1) 2009 (2) 2008

(3) 2007 (4) 1

158. In a game of minesweeper, a number on a square denotes the number of mines that share at least one vertex with that square. A square with a number may not have a mine, and the blank squares are undetermined. In how many ways can the mines be placed in the given configuration on blank squares?



(1) 120 (2) 105

(3) 95 (4) 100

159. A batsman can score 0, 2, 3, or 4 runs for each ball he receives. If N is the number of ways of scoring a total of 20 runs in one over of six balls, then N is divisible by

(1) 5 (2) 7

(3) 14 (4) 16

160. The coefficient of x^{50} in the expansion of $\sum_{k=0}^{100} {}^{100}C_k (x-2)^{100-k} 3^k$ is also equal to

(1) Number of ways in which 50 identical books can be distributed in 100 students, if each student can get atmost one book

(2) Number of ways in which 100 different white balls and 50 identical red balls can be arranged in a circle, if no two red balls are together

(3) Number of dissimilar terms in $(x_1 + x_2 + x_3 + ... + x_{50})^{51}$.

(4) $\frac{2 \cdot 6 \cdot 10 \cdot 14 .....198}{50!}$

161. Consider the letters of the word MATHEMATICS. Set of repeating letters = {M, A, T}, set of non-repeating letters = {H, E, I, C, S}

Column-I	**Column-II**
(A) The number of words taking all letters of the given word such that at least one repeating letter is at odd position is	(P) $28 \cdot (7!)$
(B) The number of words formed taking all letters of the given word in which no two vowels are together is	(Q) $\frac{(11)!}{(2!)^3}$
(C) The number of words formed taking all letters of the given word such that in each word both M's are together and both T's are together but both A's are not together is	(R) $210(7!)$
(D) The number of words formed taking all letters of the given word such that relative order of vowels and consonants does not change is	(S) $840(7!)$
	(T) $\frac{4!7!}{(2!)^3}$

162. There are 10 girls and 8 boys in a classroom including Mr. Ravi, Ms. Rani and Ms. Radha. A list of speakers consisting of 8 girls and 6 boys has to be prepared. Mr. Ravi refuses to speak if Ms. Rani is a speaker. Ms. Rani refuses to speak if Ms. Radha is a speaker. The number of ways the list can be prepared is a 3 digit number $n_1n_2n_3$, then $|n_3 + n_2 - n_1|$ =

163. There are n persons sitting around a circular table. They start singing a 2 minute song in pairs such that no two persons sitting together will sing together. This process is continued for 28 minutes. Find n.

164. The number of ways in which $2n$ objects of one type, $2n$ of another type and $2n$ of a third type can be divided between 2 persons so that each may have $3n$ objects is $\alpha n^2 + \beta n + \gamma$. Find the value of $(\alpha + \beta + \gamma)$.

165. Let $N = 2^{1224} - 1$, $\alpha = 2^{153} + 2^{77} + 1$ and $\beta = 2^{408} - 2^{204} + 1$. Then which of the following statement is correct?

(1) α divides N but β does not

(2) β divides N but α does not

(3) α and β both divide N

(4) Neither α nor β divides N

166. If $\alpha_1, \alpha_2, ..., \alpha_n$ are the n; n^{th} roots of unity, $\alpha_r = e^{\frac{i2(r-1)\pi}{n}}$, $r = 1, 2, ... n$ then ${}^nC_1\alpha_1 + {}^nC_2\alpha_2 + ... + {}^nC_n\alpha_n$ is equal to

(1) $\left(1+\frac{\alpha_2}{\alpha_1}\right)^n - 1$

(2) $\frac{\alpha_1}{2}[(1+\alpha_1)^n - 1]$

(3) $\frac{\alpha_1 + \alpha_{n-1} - 1}{2}$

(4) $(\alpha_1 + \alpha_{n-1})^n - 1$

167 If $\sum_{r=0}^{n}\left(\frac{r^3+2r^2+3r+2}{(r+1)^2}\right){}^nC_r = \frac{2^4+2^3+2^2-2}{3}$ then the value of n is

(1) 2 (2) 2^2

(3) 2^3 (4) 2^4

168. If ${}^{100}C_6 + 4.\ {}^{100}C_7 + 6.\ {}^{100}C_8 + 4.\ {}^{100}C_9 + {}^{100}C_{10}$ has the value equal to ${}^xC_y$; then the possible value(s) of $x + y$ can be

(1) 112 (2) 114

(3) 196 (4) 198

169. Match Column-I with Column-II.

Column-I	Column-II
(A) If the sum of first 84 terms of the series $\frac{4+\sqrt{3}}{1+\sqrt{3}}+\frac{8+\sqrt{15}}{\sqrt{3}+\sqrt{5}}+\frac{12+\sqrt{35}}{\sqrt{5}+\sqrt{7}}+\ldots..$ is $549k$, then k is equal to	(P) 3
(B) If $x, y \in R$, $x^2 + y^2 - 6x + 8y + 24 = 0$, the greatest value of $\frac{16}{5}\cos^2\left(\sqrt{x^2+y^2}\right)-\frac{24}{5}\sin\left(\sqrt{x^2+y^2}\right)$ is	(Q) 2
(C) If $(\sqrt{3}+1)^6+(\sqrt{3}-1)^6=416$, if $xyz=[(\sqrt{3}+1)^6]$, $x, y, z \in N$, (where [.] denotes the greatest integer function), then the number of ordered triplets (x, y, z) is	(R) 5
(D) If $(1+x)(1+x^2)(1+x^4).....(1+x^{128})=\sum_{r=0}^{n}x^r$, then $\frac{n}{85}$ is equal to	(S) 9

170. In the polynomial function $f(x) = (x - 1)(x^2 - 2)(x^3 - 3) ..... (x^{11} - 11)$ the coefficient of x^{60} is

171. If $\sum_{r=0}^{3n} a_r(x-4)^r = \sum_{r=0}^{3n} A_r(x-5)^r$ and $a_k = 1\ \forall\ k \geq 2n$ and $\sum_{r=0}^{3n} d_r(x-8)^r = \sum_{r=0}^{3n} B_r(x-9)^r$ and $\sum_{r=0}^{3n} d_r(x-12)^r = \sum_{r=0}^{3n} D_r(x-13)^r$ and $d_k = 1\ \forall\ k \geq 2n$. Then find the value of $\frac{A_{2n}+D_{2n}}{B_{2n}}$.

172. A set contains $3n$ members. Let P_n be the probability that S is partitioned into 3 disjoint subsets with n members in each subset such that the three largest members of S are in different subsets. Then $\lim_{n\to\infty} P_n =$

(1) 2/7
(2) 1/7
(3) 1/9
(4) 2/9

173. Mr. A's T.V. has only 4 channels; all of them quite boring so he naturally desires to switch (change) channel after every one minute. The probability that he is back to his original channel for the first time after 4 minutes can be expressed as $\frac{m}{n}$; where m and n are relatively prime numbers. Then $(m + n)$ equals

(1) 27 (2) 31

(3) 23 (4) 33

174. Two friends visit a restaurant randomly during 5 pm to 6 pm. Among the two, whoever comes first waits for 15 min and then leaves. The probability that they meet is

(1) $\frac{1}{4}$ (2) $\frac{1}{16}$

(3) $\frac{7}{16}$ (4) $\frac{9}{16}$

175. Three numbers are randomly selected from the set {10, 11, 12, ..., 100}. Probability that they form a Geometric progression with integral common ratio greater than 1 is

(1) $\frac{15}{{}^{91}C_3}$ (2) $\frac{16}{{}^{91}C_3}$

(3) $\frac{17}{{}^{91}C_3}$ (4) $\frac{18}{{}^{91}C_3}$

176. A consignment of 15 record players contain 4 defectives. The record players are selected at random, one by one and examined. The one examined is not put back. Then

(1) Probability of getting exactly 3 defectives in the examination of 8 recrod players is $\frac{{}^4C_3 \; {}^{11}C_5}{{}^{15}C_8}$

(2) Probability that 9th one examined is the last defective is $\frac{8}{197}$

(3) Probability that 9th examined record player is defective, given that there are 3 defectives in first 8 players examined is $\frac{1}{7}$

(4) Probability that 9th one examined is the last defective is $\frac{8}{195}$

177. There are 3 different pairs (i.e., 6 units say a, a, b, b, c, c) of shoes in a lot. Now three person come and pick the shoes randomly (each gets 2 units). Let p be the probability that no one is able to wear shoes (i.e., no one gets a correct pair), then the value of $\frac{13p}{4-p}$, is

178. Mr. B has two fair 6-sided dice, one whose faces are numbered 1 to 6 and the second whose faces are numbered 3 to 8. Twice, he randomly picks one of dice (each dice equally likely) and rolls it. Given the sum of the resulting two rolls is 9. The probability he rolled same dice twice is $\frac{m}{n}$ where m and n are relatively prime positive integers. Find $(m + n)$.

179. If a, b, c are positive numbers such that $a^{\log_3 7} = 27$, $b^{\log_7 11} = 49$, $c^{\log_{11} 25} = \sqrt{11}$, then the sum of digits of $S = a^{(\log_3 7)^2} + b^{(\log_7 11)^2} + c^{(\log_{11} 25)^2}$ is

(1) 15 (2) 17

(3) 19 (4) 21

180. If $2^{2010} = a_n 10^n + a_{n-1} 10^{n-1} + ... + a_2 10^2 + a_1 . 10 + a_0$, where $a_i \in \{0, 1, 2, ..., 9\}$ for all $i = 0, 1, 2, 3, ..., n$, then $n =$

(1) 603 (2) 604

(3) 605 (4) 606

181. Product of all value of x satisfying the equation $\sqrt{2^x \sqrt[3]{4^x (0.125)^{1/x}}} = 4(\sqrt[3]{2})$ is

(1) $\frac{14}{5}$ (2) 3

(3) $-\frac{1}{5}$ (4) $-\frac{3}{5}$

182. If x_1 and x_2 are the roots of the equation $e^2 x^{\ln x} = x^3$ with $x_1 > x_2$, then

(1) $x_1 = 2x_2$ (2) $x_1 = x_2^2$

(3) $2x_1 = x_2^2$ (4) $x_1^2 = x_2^2$

183. Identify the correct option.

(1) $\log_2 3 < \log_{1/4} 5$

(2) $\log_5 7 < \log_8 3$

(3) $\log_{\sqrt[3]{2}} \sqrt{3} > \log_{\sqrt[3]{2}} \sqrt{5}$

(4) $2^{\frac{1}{4}} > \left(\frac{3}{2}\right)^{1/3}$

184. The value of $x + y + z$ satisfying the system of equations

$\log_2 x + \log_4 y + \log_4 z = 2$ is

$\log_3 y + \log_9 z + \log_9 x = 2$

$\log_4 z + \log_{16} x + \log_{16} y = 2$

(1) $\frac{175}{12}$ (2) $\frac{349}{24}$

(3) $\frac{353}{24}$ (4) $\frac{112}{3}$

Paragraph for Q. Nos. 185 and 186

Given a right triangle *ABC* right angled at *C* and whose legs are given $1+4\log_{p^2}(2p)$, $1+2^{\log_2(\log_2 p)}$ and hypotenuse is given to be 1 + $\log_2$ (4*p*). The area of ΔABC and circle circumscribing it are Δ_1 and Δ_2 respectively, then

185. $\Delta_1+\frac{4\Delta_2}{\pi}$ is equal to

(1) 31

(2) 28

(3) $3+\frac{1}{\sqrt{2}}$

(4) 199

186. The value of $\sin\left(\frac{\pi(25p^2\Delta_1+2)}{6}\right)=$

(1) $\frac{1}{2}$ (2) $\frac{1}{\sqrt{2}}$

(3) $\frac{\sqrt{3}}{2}$ (4) 1

187. Match Column-I with Column-II.

Column-I	Column-II
(A) If $a=3(\sqrt{8+2\sqrt{7}}-\sqrt{8-2\sqrt{7}})$, $b=\sqrt{(42)(30)+36}$, then the value of $\log_a b$ is equal to	(P) –1
(B) If $a=(\sqrt{4+2\sqrt{3}}-\sqrt{4-2\sqrt{3}})$, $b=\sqrt{11+6\sqrt{2}}-\sqrt{11-6\sqrt{2}}$ then the value of $\log_a b$ is equal to	(Q) 1
(C) If $a=\sqrt{3+2\sqrt{2}}$, $b=\sqrt{3-2\sqrt{2}}$, then the value of $\log_a b$ is equal to	(R) 2
(D) If $a=\sqrt{7+\sqrt{7^2-1}}$, $b=\sqrt{7-\sqrt{7^2-1}}$, then the value of $\log_a b$ is equal to	(S) $\frac{3}{2}$
	(T) None of these

188. Let a, b, c, d are positive integers such that $\log_a b = \frac{3}{2}$ and $\log_c d = \frac{5}{4}$. If $(a - c) = 9$. Find the value of $(b - d)$.

189. The ordered pair (x, y) satisfying the equation

$$x^2 = 1 + 6\log_4 y \text{ and } y^2 = 2^x y + 2^{2x+1}$$

are (x_1, y_1) and (x_2, y_2), then find the value of $\log_2 | x_1 x_2 y_1 y_2 |$.

190. The complete set of real values of a such that the point $P(a, \sin a)$ lies inside the triangle formed by the lines $x - 2y + 2 = 0$; $x + y = 0$ and $x - y - \pi = 0$, is

(1) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$

(2) $\left(\frac{\pi}{2}, \pi\right) \cup \left(\frac{2\pi}{2}, 2\pi\right)$

(3) $(0, \pi)$

(4) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$

191. A piece of cheese is located at (12, 10) in a coordinate plane. A mouse is at (4, –2) and is running up the line $y = -5x + 18$. At the point (a, b), the mouse starts getting farther from the cheese rather than closer to it. The value of $(a + b)$ is

(1) 6 (2) 10

(3) 18 (4) 14

192. If the equation $4y^3 - 8a^2yx^2 - 3ay^2x + 8x^3 = 0$ represent three straight lines, two of them are perpendicular then sum of all possible values of a is equal to

(1) $\frac{3}{8}$ (2) $\frac{-3}{4}$

(3) $\frac{1}{4}$ (4) –2

193. The equations of L_1 and L_2 are $y = mx$ and $y = nx$, respectively. Suppose L_1 make twice as large of an angle with the horizontal (measured counterclockwise from the positive x-axis) as does L_2 and that L_1 has 4 times the slope of L_2. If L_1 is not horizontal, then the value of the product (mn) equals

(1) $\frac{\sqrt{2}}{2}$ (2) $-\frac{\sqrt{2}}{2}$

(3) 2 (4) –2

194. The minimum value of $[(x_1 - x_2)^2 + (12 - \sqrt{1 - x_1^2} - \sqrt{4x_2})^2]^{1/2}$ for all permissible value of x_1 and x_2 is equal to $a\sqrt{b} - c$ where $a, b, c \in N$, then find the value of $a + b - c$.

195. Let P, Q, R, S be the feet of the perpendiculars drawn from a point (1, 1) upon the lines $x + 4y = 12$; $x - 4y + 4 = 0$ and their angle bisectors respectively; then equation of the circle which passes through Q, R, S is

(1) $x^2 + y^2 - 5x + 3y - 6 = 0$ (2) $x^2 + y^2 - 5x - 3y + 6 = 0$

(3) $x^2 + y^2 - 5x - 3y - 6 = 0$ (4) None of these

196. Let all the points on the curve $x^2 + y^2 - 10x = 0$ are reflected about the line $y = x + 3$. The locus of the reflected points is in the form $x^2 + y^2 + gx + fy + c = 0$. The value of $(g + f + c)$ is equal to

(1) 28

(2) –28

(3) 38

(4) –38

197. A square *OABC* is formed by line pairs $xy = 0$ and $xy + 1 = x + y$ where *O* is the origin. A circle with centre C_1 inside the square is drawn to touch the line pair $xy = 0$ and another circle with centre C_2 and radius twice that of C_1, is drawn to touch the circle C_1 and the other line pair. The radius of the circle with centre C_1 is

(1) $\frac{\sqrt{2}}{\sqrt{3}(\sqrt{2}+1)}$ (2) $\frac{2\sqrt{2}}{3(\sqrt{2}+1)}$

(3) $\frac{\sqrt{2}}{3(\sqrt{2}+1)}$ (4) $\frac{\sqrt{2}+1}{3\sqrt{2}}$

198. The circle with equation $x^2 + y^2 = 1$ intersects the line $y = 7x + 5$ at two distinct points *A* and *B*. Let *C* be the point at which the positive *x*-axis intersects the circle. The angle *ACB* is

(1) $\tan^{-1}\frac{4}{3}$ (2) $\cot^{-1}(-1)$

(3) $\tan^{-1} 1$ (4) $\cot^{-1}\frac{4}{3}$

199. A circle of radius unity is centered at origin. Two particles start moving at the same time from the point (1, 0) and move around the circle in opposite direction. One of the particle moves counter clockwise with constant speed *v* and the other moves clockwise with constant speed 3*v*. After leaving (1, 0), the two particles meet first at a point *P*, and continue until they meet next at point *Q*. The coordinates of the point *Q* are

(1) (1, 0) (2) (0, 1)

(3) (0, –1) (4) (–1, 0)

Paragraph for Q. Nos. 200 and 201

Let the diameter of a subset *S* of the plane be defined as the maximum of the distance between arbitrary pairs of points of *S*.

200. Let $S = \{(x, y) : (y - x) \le 0,\ x + y \ge 0,\ x^2 + y^2 \le 2\}$, then the diameter of *S* is

(1) 2 (2) 4

(3) $\sqrt{2}$ (4) $2\sqrt{2}$

201. Let $S = \{(x, y) : (\sqrt{5}-1)x - \sqrt{10+2\sqrt{5}}\,y \ge 0,\ (\sqrt{5}-1)x + \sqrt{10+12\sqrt{5}}\,y \ge 0,\ x^2 + y^2 \le 9\}$ then the diameter of *S* is

(1) $\frac{3}{2}(\sqrt{5}-1)$ (2) $3(\sqrt{5}-1)$

(3) $3\sqrt{2}$ (4) 3

202. Match Column-I with Column-II.

Column-I	Column-II
(A) A ray of light coming from the point (1, 2) is reflected at a point A on the x-axis then passes through the point (5, 3). The coordinates of the point A are	(P) $\left(\frac{13}{5}, 0\right)$
(B) The equation of three sides of triangle ABC are $x + y = 3$, $x - y = 5$ and $3x + y = 4$. Considering the sides as diameter, three circles S_1, S_2, S_3 are drawn whose radical centre is at	(Q) (4, –1)
(C) It the straight line $x - 2y + 1 = 0$ intersects the circle $x^2 + y^2 = 25$ at the points P and Q, then the coordinate of the point of intersection of tangents drawn at P and Q to the circle is	(R) (–25, 50)
(D) The equation of three sides of a triangle are $4x + 3y + 9 = 0$, $2x + 3 = 0$ and $3y - 4 = 0$. The circum centre of the triangle is	(S) $\left(\frac{-19}{8}, \frac{1}{6}\right)$
	(T) (–1, 2)

203. Let $S = \{(x, y) \mid x, y \in R, x^2 + y^2 - 10x + 16 = 0\}$. The largest value of $\frac{y}{x}$ can be put in the form $\frac{m}{n}$ where m, n are relatively prime natural numbers, then $m^2 + n^2 =$

204. A trapezium is inscribed in the parabola $y^2 = 4x$, such that its diagonal pass through the point (1, 0) and each has length $\frac{25}{4}$. If the area of the trapezium be P, then $4P$ is equal to

(1) 70 (2) 71

(3) 80 (4) 75

205. Let P and Q are points on the parabola $y^2 = 4ax$ with vertex O, such that OP is perpendicular to OQ and have lengths r_1 and r_2 respectively, then the value of $\frac{r_1^{4/3} r_2^{4/3}}{r_1^{2/3} + r_2^{2/3}}$ is

(1) $16a^2$ (2) a^2

(3) $4a$ (4) $8a$

206. The chord AC of the parabola $y^2 = 4ax$ subtends an angle of 90º at points B and D on the parabola. If points A, B, C and D are represented by $(at_i^2, 2at_i)$, i = 1, 2, 3, 4 respectively, then find the value of $\left|\frac{t_2 + t_4}{t_1 + t_3}\right|$.

207. If lines $2x + 3y = 10$ and $2x - 3y = 10$ are tangents at the extremities of a latus rectum of an ellipse; whose centre is origin, then the length of the latus rectum is

(1) $\frac{110}{27}$ (2) $\frac{98}{27}$

(3) $\frac{100}{27}$ (4) $\frac{120}{27}$

Paragraph for Q. Nos. 208 to 210

A coplanar beam of light emerging from a point source have the equation $\lambda x - y + 2(1 + \lambda) = 0$, $\forall\ \lambda \in R$; the rays of the beam strike an elliptical surface and get reflected inside the ellipse. The reflected rays form another convergent beam having the equation $\mu x - y + 2(1 - \mu) = 0$, $\forall\ \mu \in R$. Further it is found that the foot of the perpendicular from the point (2, 2) upon any tangent to the ellipse lies on the circle $x^2 + y^2 - 4y - 5 = 0$

208. The eccentricity of the ellipse is equal to

(1) $\frac{1}{3}$ (2) $\frac{1}{\sqrt{3}}$

(3) $\frac{2}{3}$ (4) $\frac{1}{2}$

209. The area of the largest triangle that an incident ray and corresponding reflected ray can enclose with the major axis of the ellipse is equal to

(1) $4\sqrt{5}$ (2) $\sqrt{5}$

(3) $3\sqrt{5}$ (4) $2\sqrt{5}$

210. The least value of total distance travelled by an incident ray and the corresponding reflected ray is equal to

(1) 6 (2) 3

(3) $\sqrt{5}$ (4) $2\sqrt{5}$

211. Let $PQ : 2x + y + 6 = 0$ is a chord of the curve $x^2 - 4y^2 = 4$. Coordinates of the point $R(\alpha, \beta)$ that satisfy $\alpha^2 + \beta^2 - 1 \le 0$; such that area of triangle PQR is minimum; are given by

(1) $\left(\frac{-2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$ (2) $\left(\frac{-2}{\sqrt{5}}, \frac{-1}{\sqrt{5}}\right)$

(3) $\left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$ (4) $\left(\frac{2}{\sqrt{5}}, \frac{-1}{\sqrt{5}}\right)$

212. Let the chord $x\cos\alpha + y\sin\alpha = p$ of the hyperbola $\frac{x^2}{16} - \frac{y^2}{18} = 1$ subtends a right angle at the centre. Let diameter of the circle, concentric with the hyperbola, to which the given chord is a tangent is d, then $\frac{d}{4}$ is equal to

(1) 4 (2) 5

(3) 6 (4) 7

213. Let $f(x) = 2\operatorname{cosec} 2x + \sec x + \operatorname{cosec} x$, then minimum value of $f(x)$ for $x \in \left(0, \frac{\pi}{2}\right)$ is

(1) $\frac{1}{\sqrt{2}-1}$ (2) $\frac{2}{\sqrt{2}-1}$

(3) $\frac{1}{\sqrt{2}+1}$ (4) $\frac{2}{\sqrt{2}+1}$

214. Let $0 \le \alpha, \beta, \gamma, \delta \le \pi$ where β and γ are not complementary such that

$2\cos\alpha + 6\cos\beta + 7\cos\gamma + 9\cos\delta = 0$

and $2\sin\alpha - 6\sin\beta + 7\sin\gamma - 9\sin\delta = 0$

If $\frac{\cos(\alpha+\delta)}{\cos(\beta+\gamma)} = \frac{m}{n}$ where m and n are relatively prime positive numbers, then the value of $(m + n)$ is equal to

(1) 11 (2) 10

(3) 9 (4) 7

215. The number of real values of x such that

$(2^x + 2^{-x} - 2\cos x)(3^{x+\pi} + 3^{-x-\pi} + 2\cos x)(5^{\pi-x} + 5^{x-\pi} - 2\cos x) = 0$ is

(1) 1 (2) 2

(3) 3 (4) Infinite

216. If $f(n) = \prod_{r=1}^{n} \cos r,\ n \in N$, then

(1) $|f(n)| > |f(n+1)|$ (2) $f(5) > 0$

(3) $f(4) > 0$ (4) $|f(n)| < |f(n+1)|$

217. If $P = (\tan(3^{n+1}\theta) - \tan\theta)$ and $Q = \sum_{r=0}^{n} \frac{\sin(3^r\theta)}{\cos(3^{r+1}\theta)}$, then

(1) $P = 2Q$ (2) $P = 3Q$

(3) $2P = Q$ (4) $3P = Q$

218. If $2a = 2\tan 10^\circ + \tan 50^\circ$; $2b = \tan 20^\circ + \tan 50^\circ$

$2c = 2\tan 10^\circ + \tan 70^\circ$; $2d = \tan 20^\circ + \tan 70^\circ$

Then which of the following is/are correct?

(1) $a + d = b + c$ (2) $a + b = c$

(3) $a > b < c > d$ (4) $a < b < c < d$

219. The range of $y = \frac{\sin 4x - \sin 2x}{\sin 4x + \sin 2x}$ satisfies

(1) $y \in \left(-\infty, \frac{1}{3}\right)$ (2) $y \in \left(\frac{1}{3}, 1\right)$

(3) $y \in (1, 3)$ (4) $y \in (3, \infty)$

220. Match Column-I with Column-II.

Column-I	Column-II
(A) If $(1 + \tan 5^\circ)(1 + \tan 10^\circ).....(1 + \tan 45^\circ) = 2^{k+1}$ then k equals	(P) 0
(B) Sum of positive integral values of a for which $a^2 - 6\sin x - 5a \le 0\ \forall\ x \in R$ is	(Q) 2
(C) The minimum value of $\dfrac{\left(a+\frac{1}{a}\right)^4 - \left(a^4+\frac{1}{a^4}\right) - 2}{\left(a+\frac{1}{a}\right)^2 + a^2 + \frac{1}{a^2}}$ is	(R) 5
(D) Number of real roots of the equation $\sum_{k=1}^{3}(x-k)^2 = 0$ is	(S) 4
	(T) 5

221. Given that for $a, b, c, d \in R$, if $a\sec(200^\circ) - c\tan(200^\circ) = d$ and $b\sec(200^\circ) + d\tan(200^\circ) = c$, then find the value of $\left(\dfrac{a^2+b^2+c^2+d^2}{bd-ac}\right)\sin 20^\circ$.

222. Let α be the smallest integral value of x, $x > 0$ such that $\tan 19x = \dfrac{\cos 96^\circ + \sin 96^\circ}{\cos 96^\circ - \sin 96^\circ}$. The last digit of α is

223. The general solution of the equation $\tan^2(x + y) + \cot^2(x + y) = 1 - 2x - x^2$ lie on the line is

(1) $x = -1$ (2) $x = -2$

(3) $y = -1$ (4) $y = -2$

224. General solution of $4\sin^2 x + \tan^2 x + \operatorname{cosec}^2 x + \cot^2 x - 6 = 0$ is

(1) $n\pi \pm \dfrac{\pi}{4}$ (2) $2n\pi \pm \dfrac{\pi}{4}$

(3) $n\pi \pm \dfrac{\pi}{3}$ (4) $n\pi - \dfrac{\pi}{6}$

[Where $n \in I$]

225. The total number of ordered pairs (x, y) satisfying $|x| + |y| = 2$ and $\sin\left(\dfrac{\pi x^2}{3}\right) = 1$ is

(1) 2 (2) 4

(3) 6 (4) 8

226. Let $f(\theta)=\left(\cos\theta-\cos\frac{\pi}{8}\right)\left(\cos\theta-\cos\frac{3\pi}{8}\right)\left(\cos\theta-\cos\frac{5\pi}{8}\right)\left(\cos\theta-\cos\frac{7\pi}{8}\right)$ then

(1) Maximum value of $f(\theta)\ \forall\ \theta\in R$ is $\frac{1}{4}$

(2) Maximum value of $f(\theta)\ \forall\ \theta\in R$ is $\frac{1}{8}$

(3) $f(0)=\frac{1}{8}$

(4) Number of principle solutions of $f(\theta)=0$ is 8

227. Match Column-I with Column-II.

Column-I	Column-II
(A) If $x, y\in[0, 2\pi]$, then total number of ordered pair (x, y) satisfying $\sin x\cos y = 1$ is	(P) 4
(B) If $f(x) = \sin x - \cos x - kx + b$ decreases for all real values of x, then $2\sqrt{2}k$ may be	(Q) 0
(C) The number of solution of the equation $\sin^{-1}(\lvert x^2-1\rvert)+\cos^{-1}(\lvert 2x^2-5\rvert)=\frac{\pi}{2}$ is	(R) 2
(D) The number of ordered pair (x, y) satisfying the equation $\sin x + \sin y = \sin(x + y)$ and $\lvert x\rvert + \lvert y\rvert = 1$ is	(S) 3
	(T) 6

228. If the sum of all values of θ, $0\le\theta\le 2\pi$ satisfying the equation $(8\cos 4\theta-3)(\cot\theta+\tan\theta-2)(\cot\theta+\tan\theta+2)=12$ is $k\pi$, then k is equal to

229. Circumradius of a ΔABC is 3 units; let O be the circumcentre and H be the orthocentre then the value of $\frac{1}{64}(AH^2+BC^2)(BH^2+AC^2)(CH^2+AB^2)$ equals

(1) 3^4 (2) 9^3

(3) 27^6 (4) 81^4

230. $\Delta I_1I_2I_3$ is an excentral triangle of an equilateral triangle ABC such that $I_1I_2 = 4$ unit, if ΔDEF is pedal triangle of ΔABC, then $\frac{\text{Ar}(\Delta I_1I_2I_3)}{\text{Ar}(\Delta DEF)}$ =

(1) 16 (2) 4

(3) 2 (4) 1

231. If the sides a, b, c of a triangle ABC are the roots of the equation $x^3 - 13x^2 + 54x - 72 = 0$, then the value of $\frac{\cos A}{a}+\frac{\cos B}{b}+\frac{\cos C}{c}$ is equal to

(1) $\frac{61}{144}$ (2) $\frac{61}{72}$

(3) $\frac{169}{144}$ (4) $\frac{59}{144}$

232. A triangle has base 10 cm long and the base angles of 50º and 70º. If the perimeter of the triangle is $x + y \cos z^{\circ}$ where $z \in (0, 90)$ then the value of $x + y + z$ equals

(1) 60 (2) 55

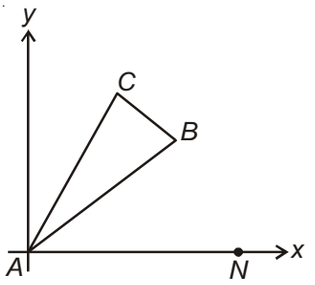
(3) 50 (4) 40

233. A is the orthocentre of ΔABC and D is reflection point of A w.r.t. perpendicular bisector of BC, then orthocentre of ΔDBC is

(1) D (2) C

(3) B (4) A

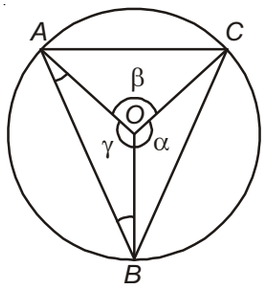
234. Find the radius of the circle escribed to the triangle ABC (shown in the figure below) on the side BC if $\angle NAB = 30^{\circ}$, $\angle BAC = 30^{\circ}$, $AB = AC = 5$.



(1) $\frac{(10\sqrt{2}+5\sqrt{3}-5)(2-\sqrt{3})}{2\sqrt{2}}$ (2) $\frac{(10\sqrt{2}+5\sqrt{3}+5)}{2\sqrt{2}}(2-\sqrt{3})$

(3) $\frac{(10\sqrt{2}+5\sqrt{3}-5)}{2\sqrt{2}}(2+\sqrt{3})$ (4) $\frac{(10\sqrt{2}+5\sqrt{2}+1)}{2\sqrt{3}}(\sqrt{3}-1)$

235. A circle of area 20 sq. units is centered at the point O. Suppose ΔABC is inscribed in the circle and has area 8 sq. units. The central angles α, β and γ are as shown in the figure. The value of $(\sin\alpha + \sin\beta + \sin\gamma)$ is equal to



(1) $\frac{4\pi}{5}$ (2) $\frac{3\pi}{4}$

(3) $\frac{2\pi}{5}$ (4) $\frac{\pi}{4}$

236. In a triangle ABC, $\tan A$ and $\tan B$ satisfy the inequality $\sqrt{3}x^2 - 4x + \sqrt{3} < 0$, then which of the following must be correct? (where symbols used have usual meanings)

(1) $a^2 + b^2 - ab < c^2$ (2) $a^2 + b^2 > c^2$

(3) $a^2 + b^2 + ab > c^2$ (4) $a^2 + b^2 < c^2$

237. In a triangle the length of two larger sides are 10 and 9 respectively. If the angles are in A.P., the length of third side can be

(1) $5-\sqrt{6}$ (2) $5+\sqrt{6}$

(3) $6-\sqrt{5}$ (4) $6+\sqrt{5}$

Paragraph for Q. Nos. 238 and 239

Let ABC be a given triangle. Points D and E are on sides AB and AC respectively and point F is on line segment DE. Let $\frac{AD}{AB}=x$, $\frac{AE}{AC}=y$, $\frac{DF}{DE}=z$. Let area of $\Delta BDF=\Delta_1$, area of $\Delta CEF=\Delta_2$ and area of $\Delta ABC=\Delta$.

238. $\frac{\Delta_1}{\Delta}$ is equal to

(1) xyz (2) $(1-x)y(1-z)$

(3) $(1-x)yz$ (4) $x(1-y)z$

239. $\frac{\Delta_2}{\Delta}$ is equal to

(1) $(1-x)y(1-z)$ (2) $(1-x)(1-y)z$

(3) $x(1-y)(1-z)$ (4) $(1-x)yz$

240. In an acute angled triangle ABC, $\angle A=20^\circ$, let DEF be the feet of altitudes through A, B, C respectively and H is the orthocentre of ΔABC. Find $\frac{AH}{AD}+\frac{BH}{BE}+\frac{CH}{CF}$.

241. Let ΔABC be inscribed in a circle having radius unity. The three internal bisectors of the angles A, B and C are extended to intersect the circumcircle of ΔABC at A_1, B_1 and C_1 respectively. Then

$$\frac{AA_1\cos\frac{A}{2}+BB_1\cos\frac{B}{2}+CC_1\cos\frac{C}{2}}{\sin A+\sin B+\sin C}=$$

242. The solution set of $(\cot^{-1}x)(\tan^{-1}x)+\left(2-\frac{\pi}{2}\right)\cot^{-1}x-3\tan^{-1}x-3\left(2-\frac{\pi}{2}\right)>0$, is

(1) $x\in(\tan 2,\ \tan 3)$

(2) $x\in(\cot 3,\ \cot 2)$

(3) $x\in(-\infty,\ \tan 2)\cup(\tan 3,\ \infty)$

(4) $x\in(-\infty,\ \cot 3)\cup(\cot 2,\ \infty)$

243. Let α, β are the roots of the equation $x^2+7x+k(k-3)=0$, where $k\in(0,3)$ and k is a constant. Then the value of $\tan^{-1}\alpha+\tan^{-1}\beta+\tan^{-1}\frac{1}{\alpha}+\tan^{-1}\frac{1}{\beta}$ is

(1) π (2) $\frac{\pi}{2}$

(3) 0 (4) $-\frac{\pi}{2}$

244. The number of ordered pair(s) (x, y) which satisfy $y = \tan^{-1} \tan x$ and $16(x^2 + y^2) - 48\pi x + 16\pi y + 31\pi^2 = 0$, is

(1) 0 (2) 1

(3) 2 (4) 3

245. The complete solution set of the equation

$$\sin^{-1}\sqrt{\frac{1+x}{2}} - \sqrt{2-x} = \cot^{-1}(\tan\sqrt{2-x}) - \sin^{-1}\sqrt{\frac{1-x}{2}} \text{ is}$$

(1) $\left[2-\frac{\pi^2}{4}, 1\right]$ (2) $\left[1-\frac{\pi^2}{4}, 1\right]$

(3) $\left[2-\frac{\pi^2}{4}, 0\right]$ (4) $[-1, 1]$

246. If $3 \le a < 4$ then the value of $\sin^{-1}(\sin[a]) + \tan^{-1}(\tan[a]) + \sec^{-1}(\sec[a])$, where $[x]$ denotes greatest integer function less than or equal to x, is equal to

(1) 3

(2) $2\pi - 9$

(3) $2\pi - 3$

(4) $9 - 2\pi$

247. The value of x satisfying the equation $(\sin^{-1}x)^3 - (\cos^{-1}x)^3 + (\sin^{-1}x)(\cos^{-1}x)(\sin^{-1}x - \cos^{-1}x) = \frac{\pi^3}{16}$ cannot be equal to

(1) $\cos\frac{\pi}{5}$ (2) $\cos\frac{\pi}{4}$

(3) $\cos\frac{\pi}{8}$ (4) $\cos\frac{\pi}{12}$

248. Find number of solutions of the equation $\sin^{-1}(|\log_6^2(\cos x) - 1|) + \cos^{-1}(|3\log_6^2(\cos x) - 7|) = \frac{\pi}{2}$, if $x \in [0, 4\pi]$.

249. If acute angle between the line $\vec{r} = \hat{i} + 2\hat{j} + \lambda(4\hat{i} - 3\hat{k})$ and xy-plane is θ_1 and acute angle between the planes $x + 2y = 0$ and $2x + y = 0$ is θ_2, then $(\cos^2\theta_1 + \sin^2\theta_2)$ equals to

(1) 1 (2) $\frac{1}{4}$

(3) $\frac{2}{3}$ (4) $\frac{3}{4}$

250. If a, b, c, x, y, z are real and $a^2 + b^2 + c^2 = 25$, $x^2 + y^2 + z^2 = 36$ and $ax + by + cz = 30$, then $\frac{a+b+c}{x+y+z}$ is equal to

(1) 1 (2) $\frac{6}{5}$

(3) $\frac{5}{6}$ (4) $\frac{3}{4}$

251. Consider matrices $A=\begin{bmatrix}1 & 2 & 3\\ 4 & 1 & 2\\ 1 & -1 & 1\end{bmatrix}$; $B=\begin{bmatrix}2 & 1 & 3\\ 4 & 1 & -1\\ 2 & 2 & 3\end{bmatrix}$; $C=\begin{bmatrix}14\\ 12\\ 2\end{bmatrix}$; $D=\begin{bmatrix}13\\ 11\\ 14\end{bmatrix}$; $X=\begin{bmatrix}x\\ y\\ z\end{bmatrix}$ such that soutions of equation $AX = C$ and $BX = D$ represents two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ respectively in three dimensional space. If $P'Q'$ is the reflection of the line PQ in the plane $\prod: x+y+z=9$, then the point which does not lie on $P'Q'$ is

(1) (3, 4, 2) (2) (5, 3, 4)

(3) (7, 2, 3) (4) (1, 5, 6)

252. The distance between the line $\vec{r}=2\hat{i}-2\hat{j}+3\hat{k}+\lambda(\hat{i}-\hat{j}+4\hat{k})$ and the plane $\vec{r}\cdot(\hat{i}+5\hat{j}+\hat{k})=5$ is

(1) $\frac{10}{9}$ (2) $\frac{10}{3\sqrt{3}}$

(3) $\frac{3}{10}$ (4) $\frac{10}{3}$

253. Let $\vec{r}$ be position vector of variable point in cartesian plane OXY such that $\vec{r}\cdot(\vec{r}+6\vec{j})=7$ cuts the co-ordinate axes at four distinct points, then the area of the quadrilateral formed by joining these points is

(1) $4\sqrt{7}$ (2) $6\sqrt{7}$

(3) $7\sqrt{7}$ (4) $8\sqrt{7}$

254. If equation of three lines are $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$; $\frac{x}{2}=\frac{y}{1}=\frac{z}{3}$ and $\frac{x-1}{1}=\frac{2-y}{1}=\frac{z-3}{0}$, then which of the following statement(s) is/are correct?

(1) Triangle formed by the line is equilateral

(2) Triangle formed by the lines is isosceles

(3) Equation of the plane containing the lines is $x + y = z$

(4) Area of the triangle formed by the lines is $\frac{3\sqrt{3}}{2}$

255. Consider the line $L_1 : x = y = z$ and the line $L_2 : 2x + y + z - 1 = 0 = 3x + y + 2z - 2$, then

(1) The shortest distance between the two lines is $\frac{1}{\sqrt{2}}$

(2) The shortest distance between the two lines is $\sqrt{2}$

(3) Plane containing the line L_2 and parallel to line L_1 is $z - x + 1 = 0$

(4) Perpendicular distance of origin from plane containing line L_2 and parallel to line L_1 is $\frac{1}{\sqrt{2}}$

256. The volume of a right triangular prism $ABCA_1B_1C_1$ is equal to 3. If the position vectors of the vertices of the base ABC are $A(1, 0, 1)$; $B(2, 0, 0)$ and $C(0, 1, 0)$, then the position vectors of the vertex A_1 can be

(1) (2, 2, 2) (2) (0, 2, 0)

(3) (0, –2, 2) (4) (0, –2, 0)

257. Let *M*, *N*, *P* and *Q* be the mid-points of the edges *AB*, *CD*, *AC* and *BD* respectively of the tetrahedron *ABCD*. Further, *MN* is perpendicular to both *AB* and *CD* and *PQ* is perpendicular to both *AC* and *BD*. Then which of the following is/are correct?

(1) $AB = CD$ (2) $BC = DA$

(3) $AC = BD$ (4) $AN = BN$

Paragraph for Q. Nos. 258 to 260

The vertices of ΔABC are $A(2, 0, 0)$, $B(0, 1, 0)$, $C(0, 0, 2)$. Its orthocentre is *H* and circumcentre is *S*. *P* is a point equidistant from *A*, *B*, *C* and the origin *O*.

258. The *z*-coordinate of *H* is

(1) 1 (2) $\frac{1}{2}$

(3) $\frac{1}{6}$ (4) $\frac{1}{3}$

259. The *y*-coordinate of *S* is

(1) $\frac{5}{6}$ (2) $\frac{1}{3}$

(3) $\frac{1}{6}$ (4) $\frac{1}{2}$

260. *PA* is equal to

(1) 1 (2) $\sqrt{2}$

(3) $\sqrt{\frac{3}{2}}$ (4) $\frac{3}{2}$

261. Let *OA*, *OB*, *OC* be coterminous edges of a cuboid. If *l*, *m*, *n* be the shortest distances between the sides *OA*, *OB*, *OC* and their respective skew body diagonals to them, respectively, then find $\frac{\left(\frac{1}{l^2}+\frac{1}{m^2}+\frac{1}{n^2}\right)}{\left(\frac{1}{OA^2}+\frac{1}{OB^2}+\frac{1}{OC^2}\right)}$.

262. Let $\vec{v}_0$ be a fixed vector and $\vec{v}_0 = \begin{bmatrix}1\\0\end{bmatrix}$. Then for $n \geq 0$ a sequence is defined $\vec{v}_{n+1} = \vec{v}_n + \left(\frac{1}{2}\right)^{n+1}\begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}^{n+1}\vec{v}_0$ then $\lim_{n\to\infty}\vec{v}_n = \begin{bmatrix}\alpha\\\beta\end{bmatrix}$. Find $\frac{\alpha}{\beta}$.

263. *ABCD* is a regular tetrahedron, *A* is the origin and *B* lies on *x*-axis. *ABC* lies in the *xy*-plane $|\overrightarrow{AB}| = 2$. Under these conditions, the number of possible tetrahedrons is

❑ ❑ ❑